

The API Economy with OpenAPI2: Best Practices for IMS Transactions and Databases

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IBM IMS

**Note: this presentation only addresses OPEN API2
Support for OPEN API3 is provided via another set of materials**

Abstract

The API Economy is a growing buzz phrase in today's computing environment, but how does that apply to existing business systems? Explore the what and how of API enabling not only simple, but also complex IMS transactions and, most recently, IMS databases.

The first part focuses on z/OS Connect EE and the IMS Service Provider for IMS transaction access along with application programming model considerations. For example, what should be done when the target transaction spawns multiple other transactions and only one can respond to the request? Understand what to do in these types of complex transaction scenarios. It will also discuss the use of the API requester from IMS applications for outbound requests and will wrap up with an overview discussion on configuring this environment for IMS high availability and resiliency.

The second part focuses on the IMS DB Service Provider and how this support can be enabled and leveraged in z/OS Connect EE to access IMS data. It will also discuss how the IMS Explorer for Development can assist in building and testing SQL queries that can then be used in the creation of an API request using the z/OS Explorer toolkit.

Topics

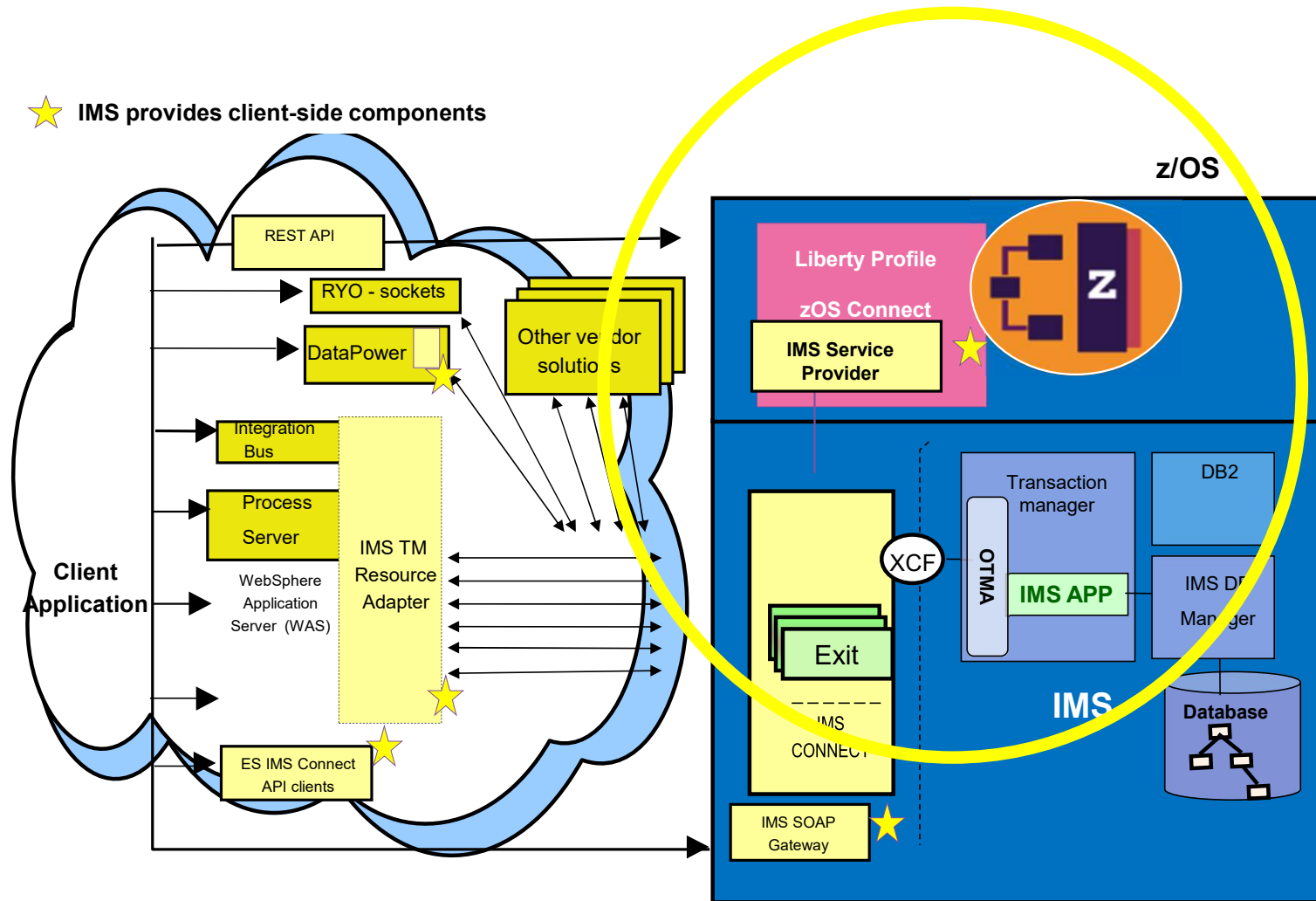
- The API economy and IMS Transactions
 - z/OS Connect EE and the IMS Service Provider
 - Is it secure?
 - How are Services and APIs created?
 - IMS implications
 - The questions to ask: Simple versus complex interactions

- High Availability and Workload balancing IMS Connect/IMS
 - Transactions (OTMA)
 - Databases (ODBM)

- The API economy and IMS Databases
 - Recently added support
 - How is it done?

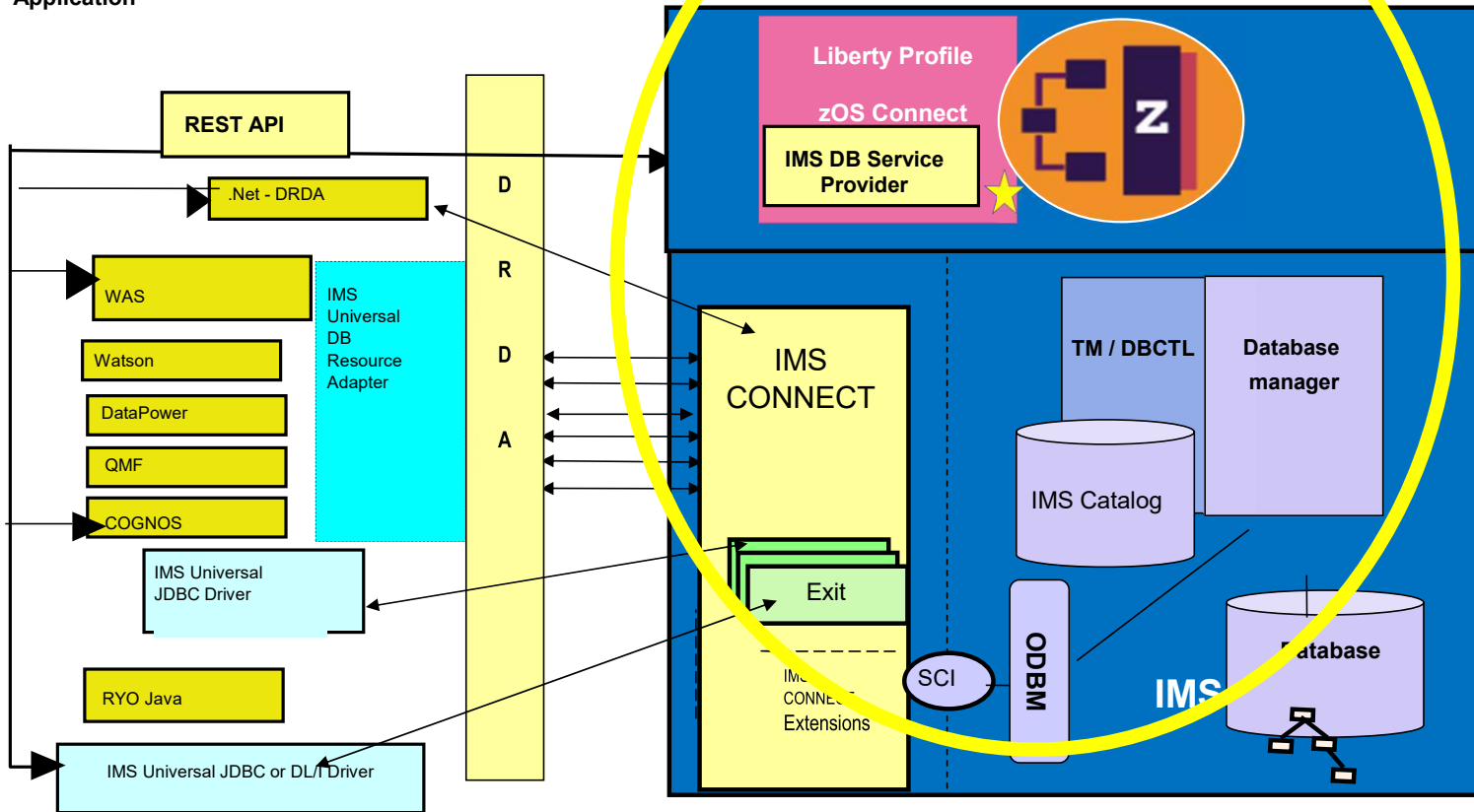
Foundation: Integration to Transactions

★ IMS provides client-side components



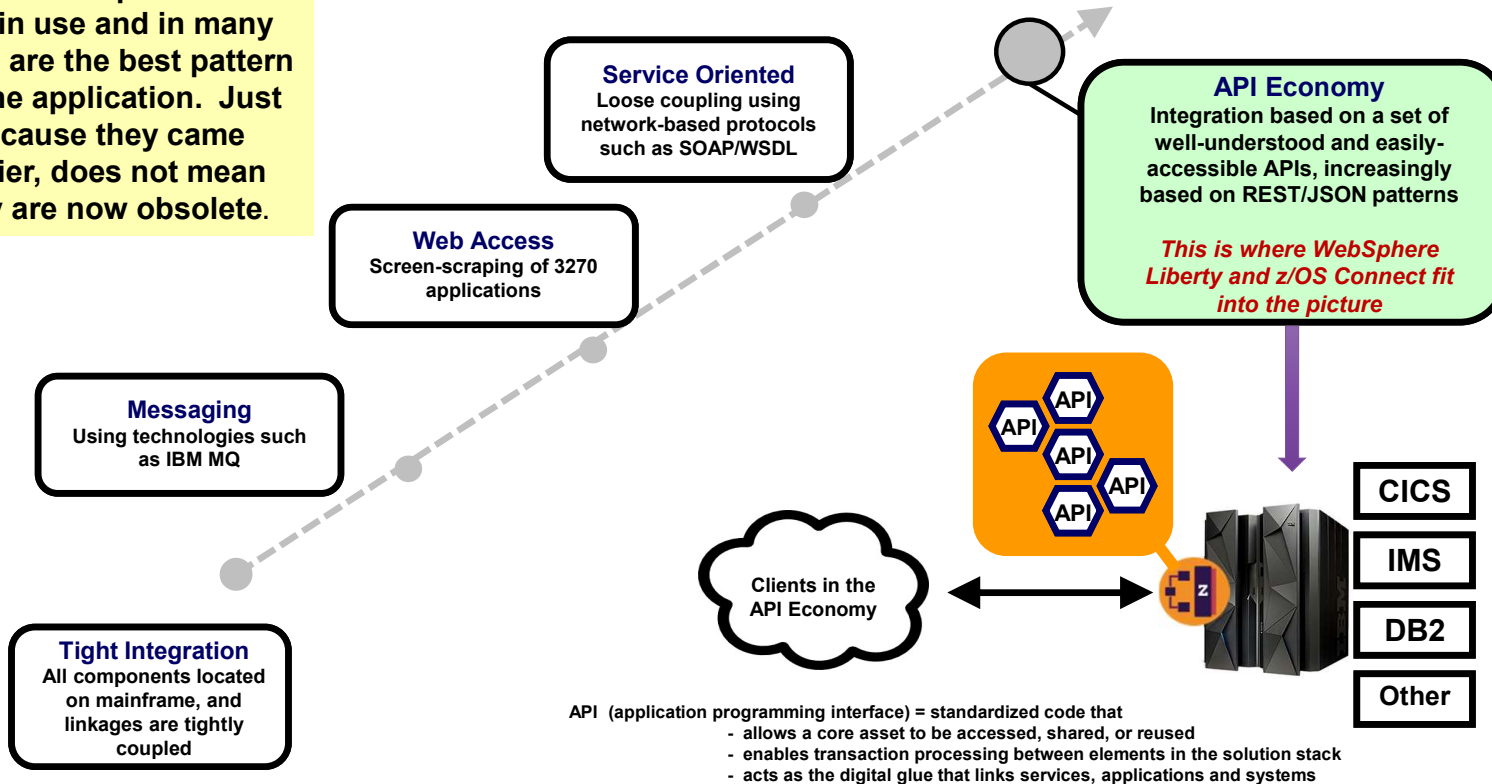
Foundation: Integration to Databases

Client
Application



The Integration Evolution

These earlier patterns are still in use and in many cases are the best pattern for the application. Just because they came earlier, does not mean they are now obsolete.



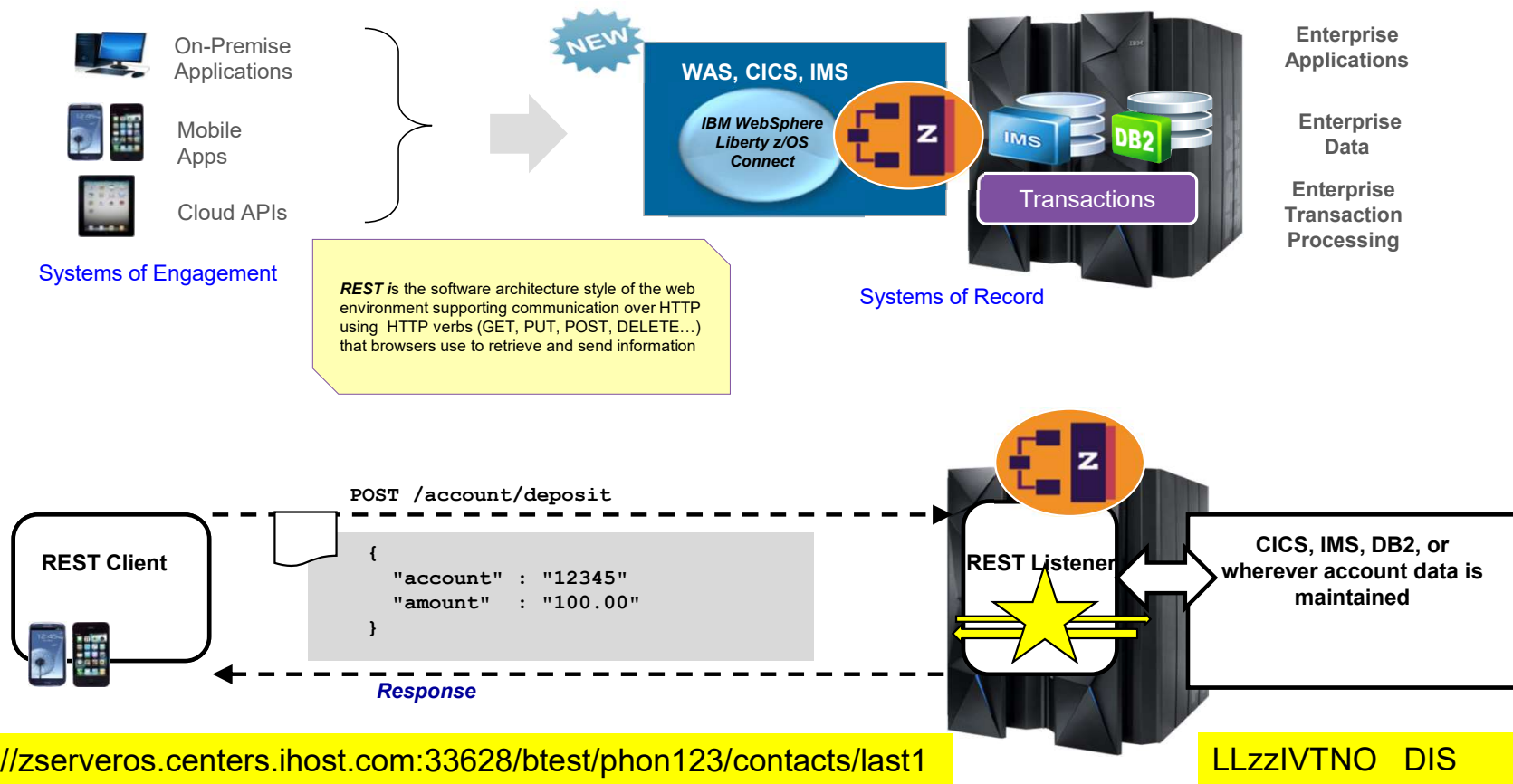
For IMS, the goal is to support the protocols best suited to your environment and needs
→ protect the application investment

Why has REST become increasingly popular as an integration pattern?

Because it is stateless, relatively lightweight, is relatively easy to program to, and operates well with discovery mechanisms such as IBM's API management solution -- API Connect.

Let's start with Transactions

For IMS: REST, JSON, z/OS Connect, and the IMS Service Provider



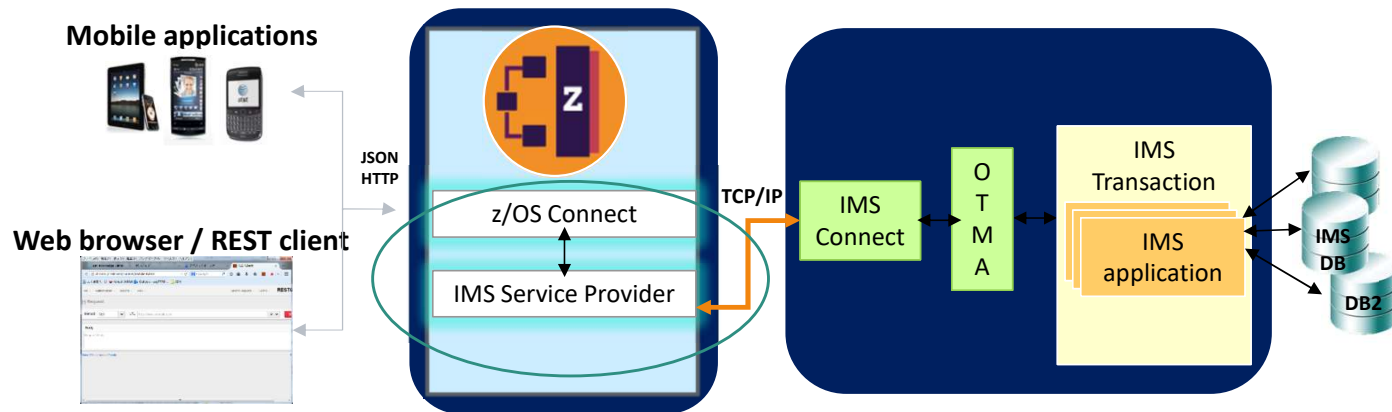
Value to IMS Environments

— **IMS Service Provider to access IMS transactions**

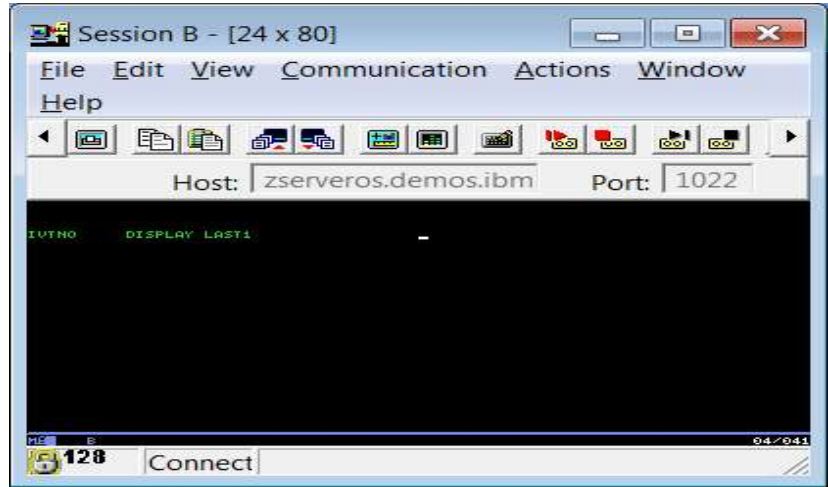
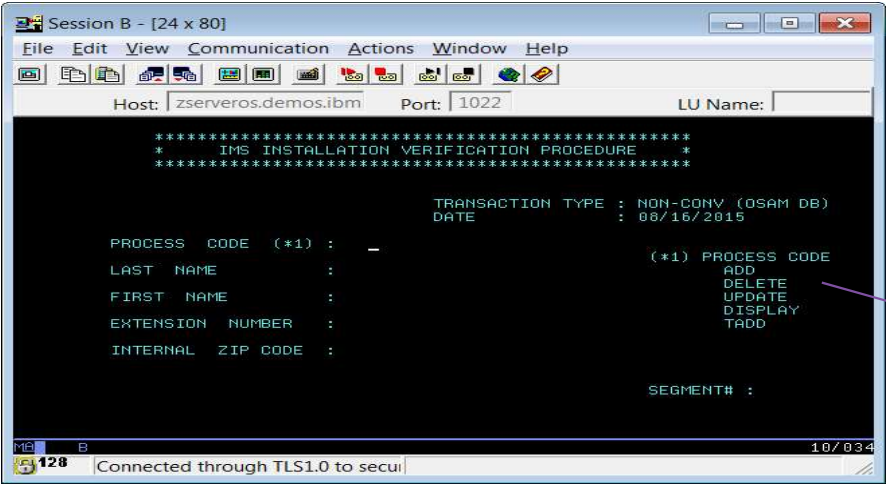
- Available with z/OS Connect EE
 - IMS provides tooling support in z/OS Explorer
- Delivers IMS resources to mobile and cloud developers in a secure, governed, and optimized way through:

An integrated platform that supports full discovery, modeling, enablement, and deployment of both IMS transactions and IMS data

- A singular approach for System z clients using WAS, CICS, IMS, and DB2

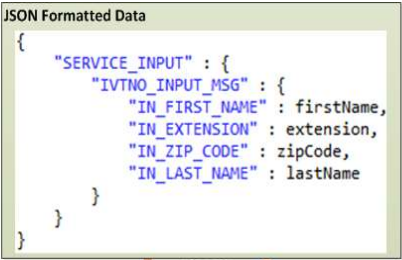
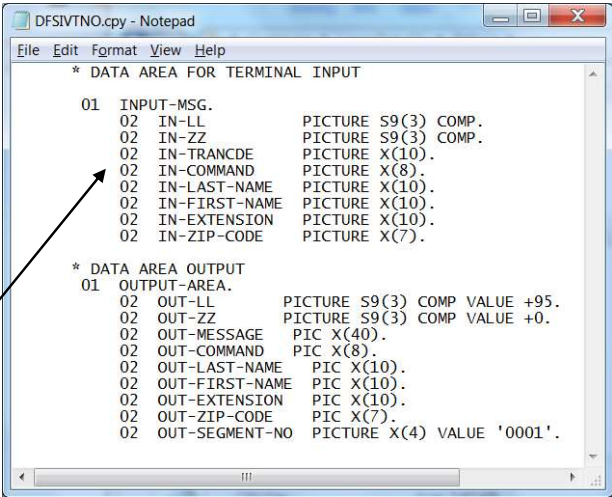


REST APIs → IMS Transactions



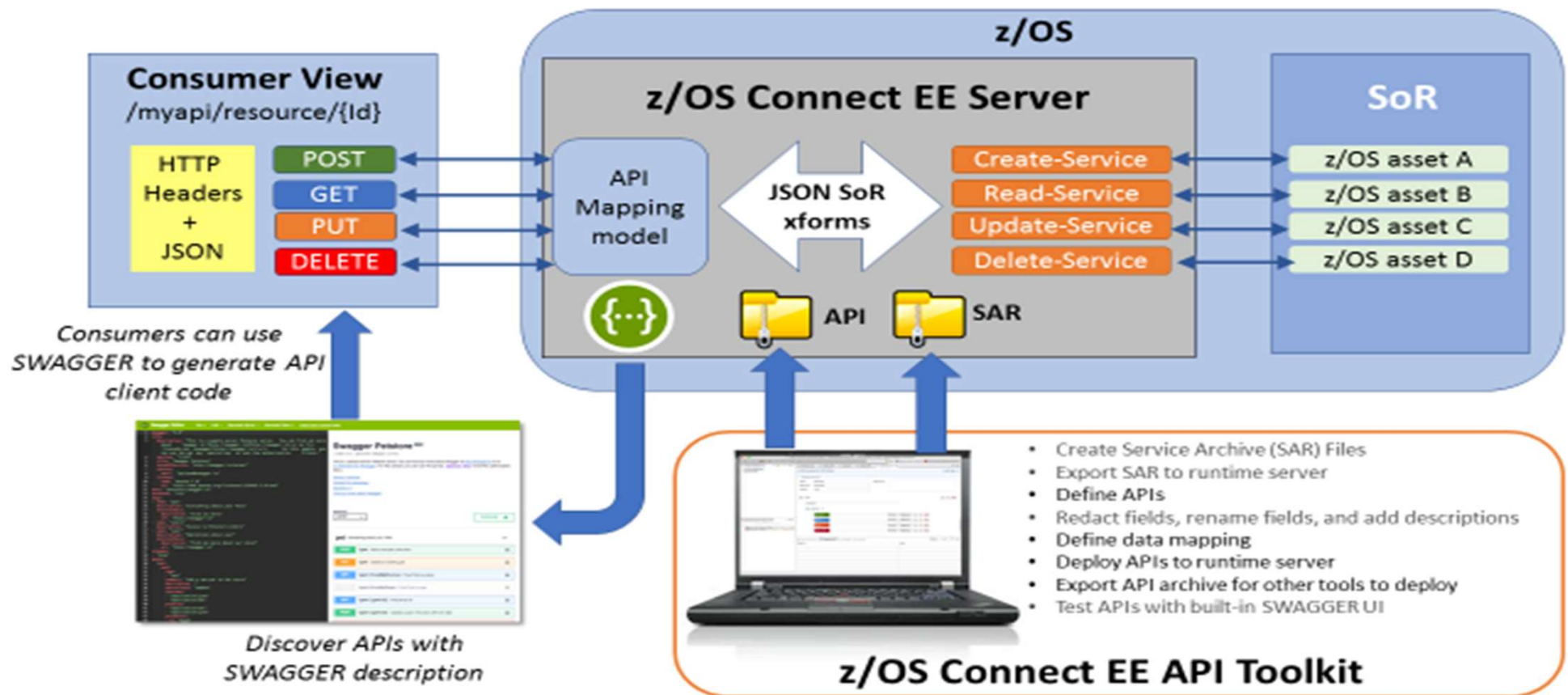
LLzZIVTNO ADD

LLzZIVTNO DIS LAST1

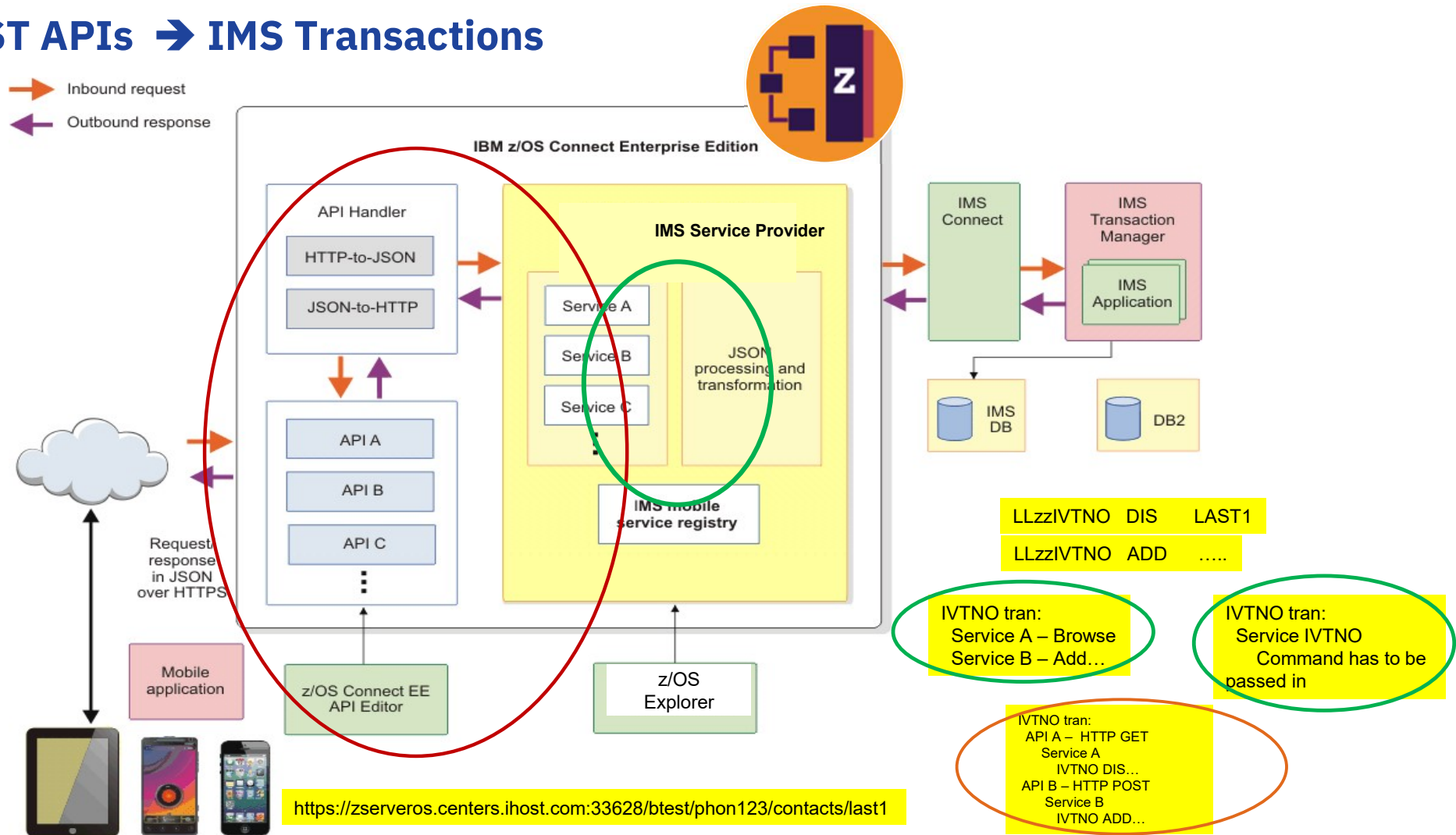


Field Name →	LL	ZZ	TRANCODE	COMMAND	LAST NAME	FIRST NAME	EXTENSION	ZIP CODE
HEX Value →	03	00	CEEDD44444	CCC44444	CDC4444444	DDCD444444	FFF6FFFF44	FFFFF44
Readable Value →	59	0	IVTNO	ADD	DOE	JOHN	555-5555	95141

Facilitated Through Tooling



REST APIs → IMS Transactions



Is it really that easy ?

It depends on the simplicity or complexity of the IMS transaction

Recommendations: always analyze the target environment for anomalies that might need to be considered

And.. Some Questions For Consideration

- **What are the security requirements**
 - *What userid can/should be used?*
 - *Is there anything in the IMS application architecture unique to the environment, e.g., application security?*
- **How simple or complex is the IMS application interaction**
 - ***Are you writing a new transaction or leveraging your existing environment?***
 - IMS conversational transactions are not supported
 - ***Are there copybooks for all the message layouts?***
 - ***Are any of the IMS transactions coded to be sensitive to the LTERMname or MODname in the IOPCB?***
 - E.g., does the application architecture simulate a conversational model, e.g., with a database that keeps information between transaction iterations?
 - ***Do any of the applications issue pgm-pgm switches (ISRT ALTPCB) to another transaction?***
 - If so, does the primary transaction also ISRT back to the IOPCB?
 - Is the architecture for switching serialized so that only one responds back to the IOPCB?
 - Are any of the pgm-pgm switches using EXPRESS ALTPCBs?
 - ***For the MFS transactions that manipulate 3270 extended attributes***
 - Are there copybooks for the messages with the layouts that include these fields?
 - Do any of these transactions have the possibility of issuing the ISRT,IOPCB using different modnames (i.e., different output message layouts)

What are the security requirements ?

- Which userid needs to be used if transaction authorization is required or the application has its own security?

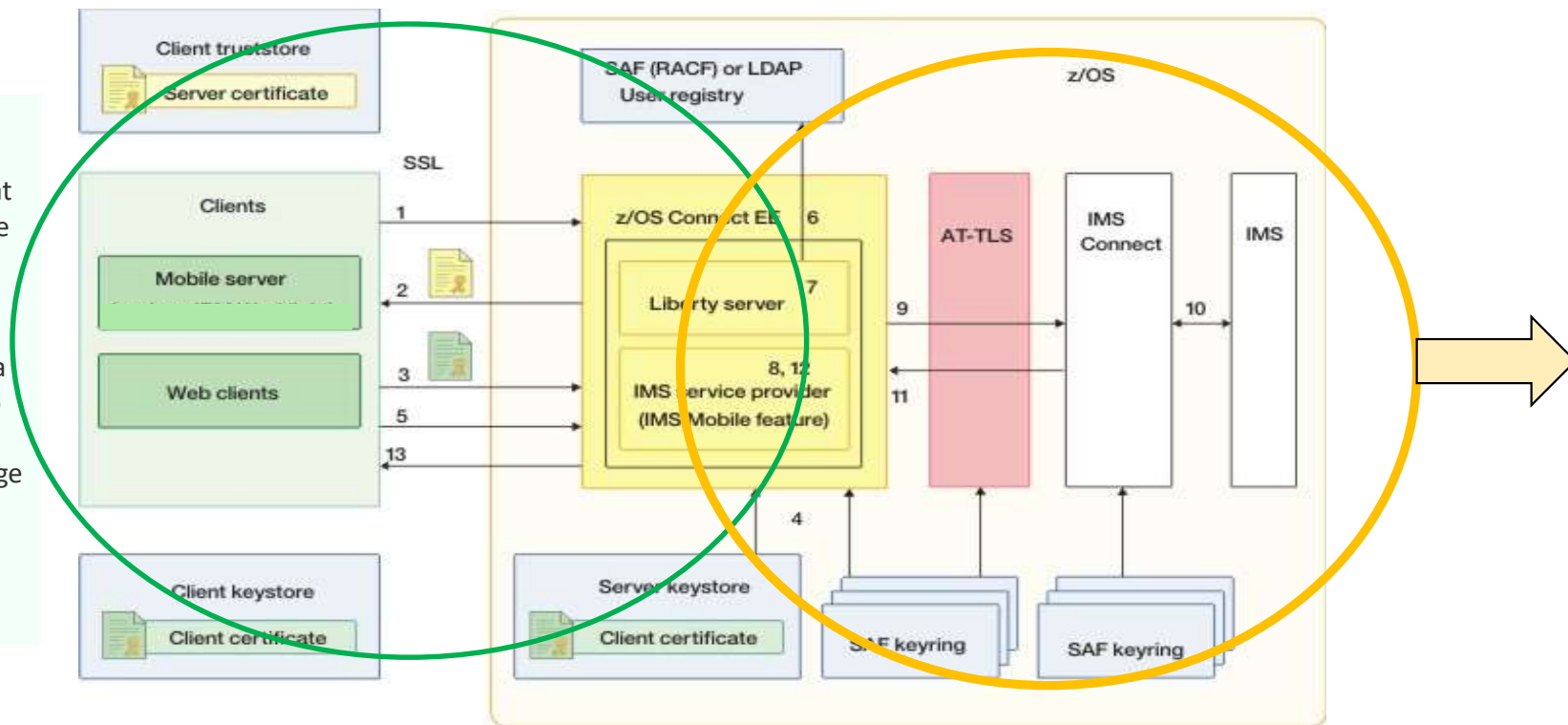
Securing the Environment

- At a... **z/OS Connect** and at an **IMS Service Provider level:**

- Basic authentication**

- Client authentication** that uses a TLS client certificate to identify the client.

- Third-party authentication** that uses a token such as a JSON Web Token (JWT), Security Assertion Markup Language (SAML) token or Lightweight Third-Party Authentication (LTPA) token. Etc.

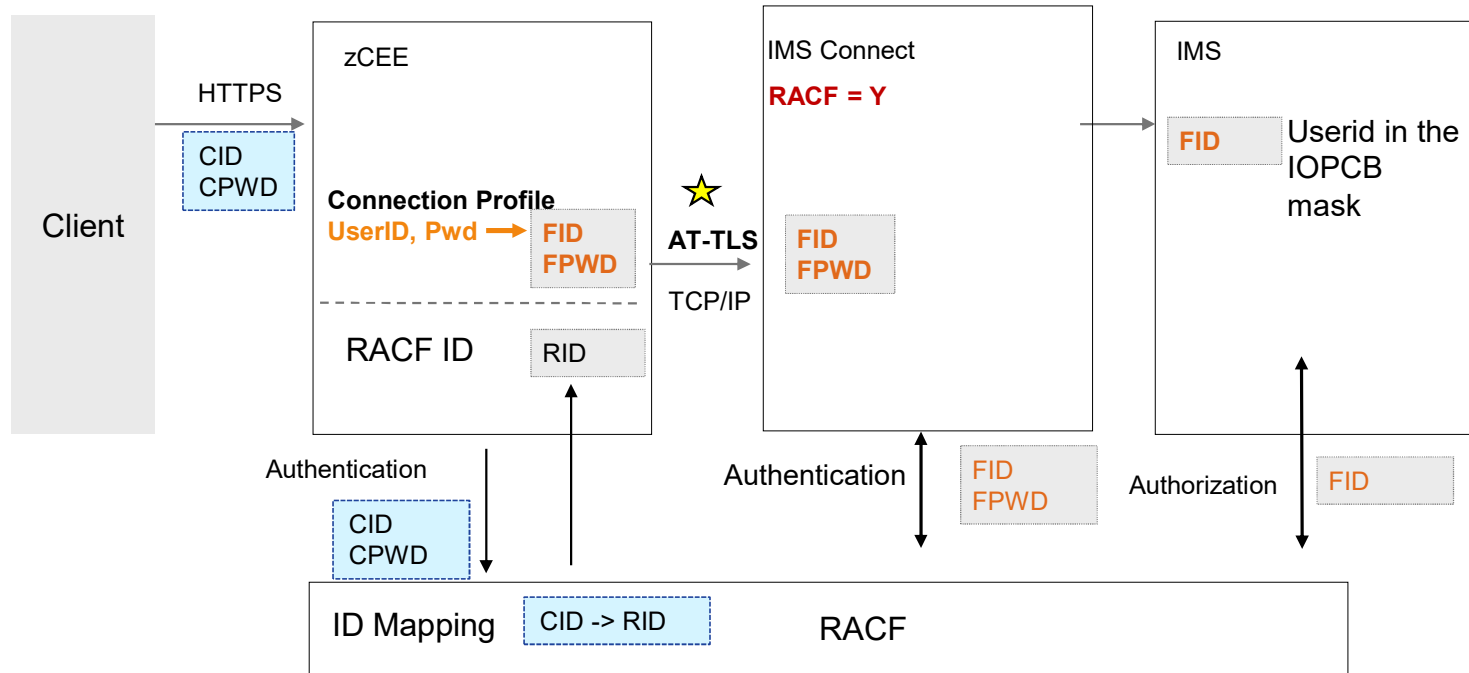


Scenario 1 – IMS SP Connection Profile Identity + RACF=Y

FID FPWD = User ID, Password specified in the connection profile

CID CPWD = User ID, Password of the originating client (Basic Auth or Cert, etc)

RID = RACF Mapped ID (No Password)

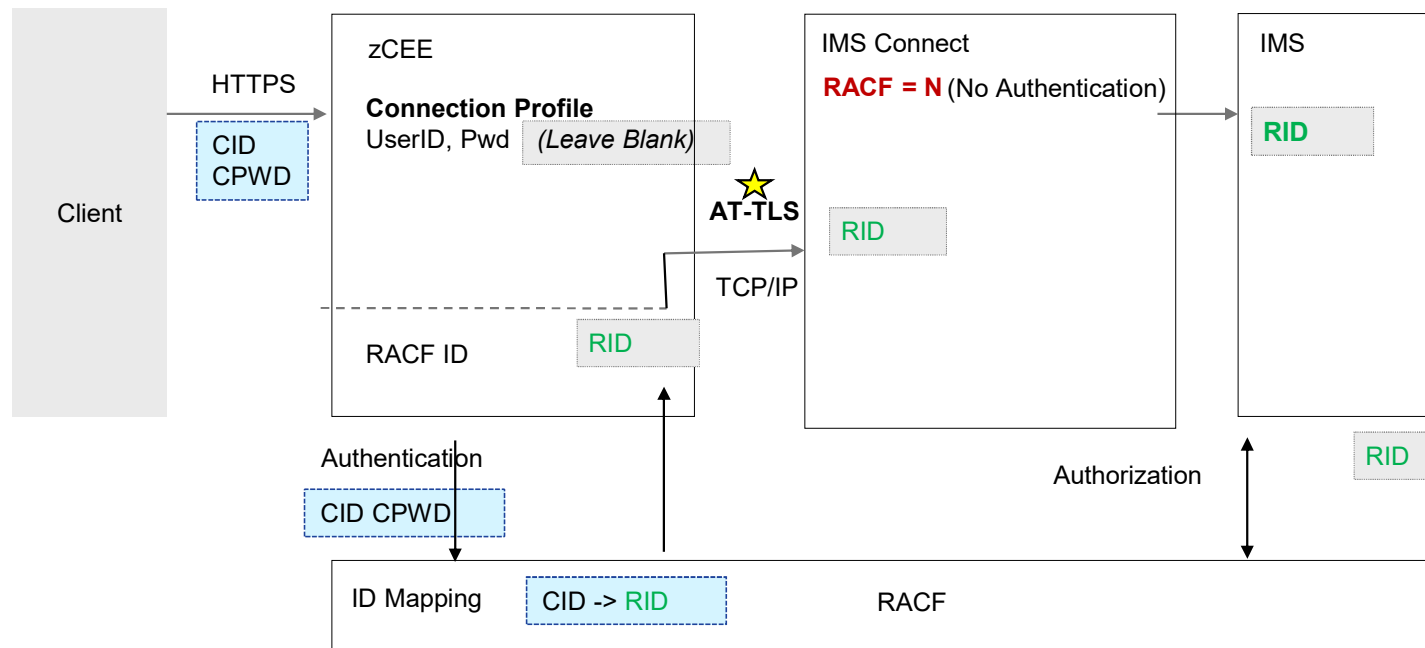


Is a generic userid (s) viable for your IMS application environment?

AT-TLS can be implemented to provide a secure connection between ZCEE and IMS Connect ★

Scenario 2 – Client Identity + RACF=N

FID FPWD = User ID, Password are NOT specified in the connection profile
CID CPWD = User ID, Password of the originating client (Basic Auth or Cert, etc)
RID = RACF Mapped ID (No Password)



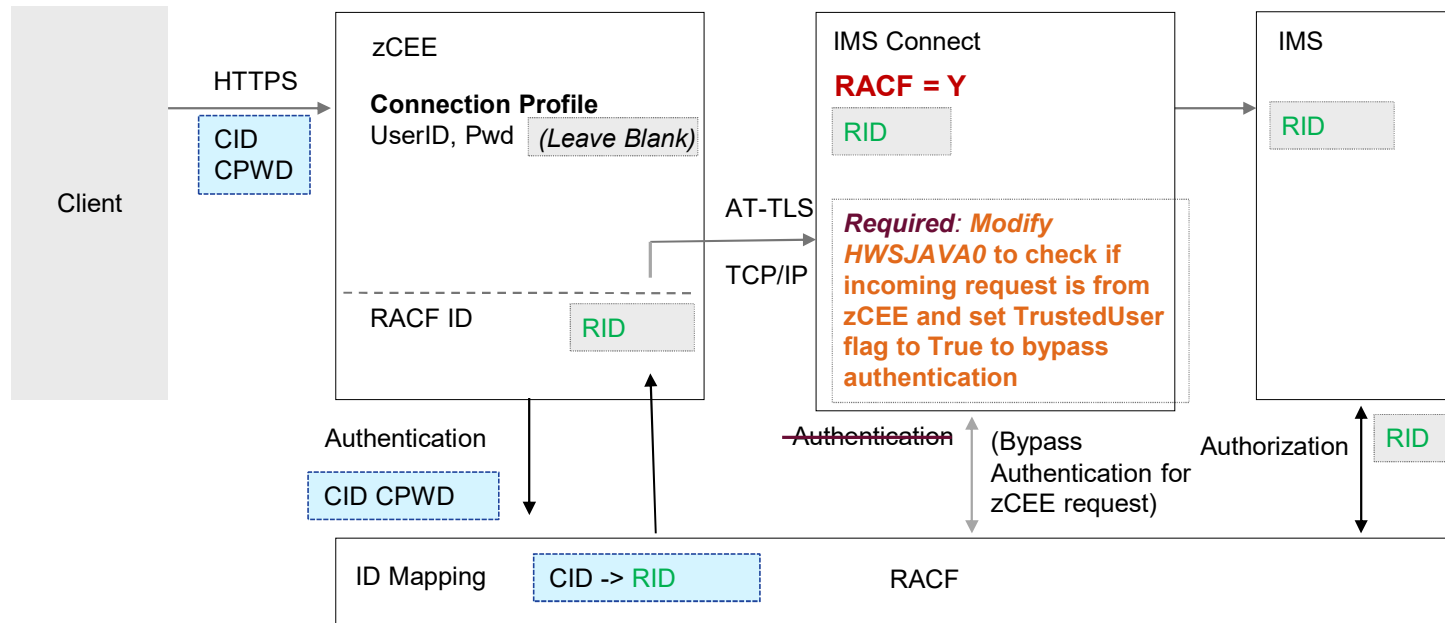
For environments that need the end-user userid to be passed into IMS....

Is bypassing authorization in IMS Connect (RACF=N) acceptable to the auditors?

- ZCEE provides security into the environment
- AT-TLS secures the connect to IMS Connect

Scenario 3 – Client Identity + RACF=Y

FID FPWD = User ID, Password are NOT specified in the connection profile
CID CPWD = User ID, Password of the originating client (Basic Auth or Cert, etc.)
RID = RACF Mapped ID (No Password)



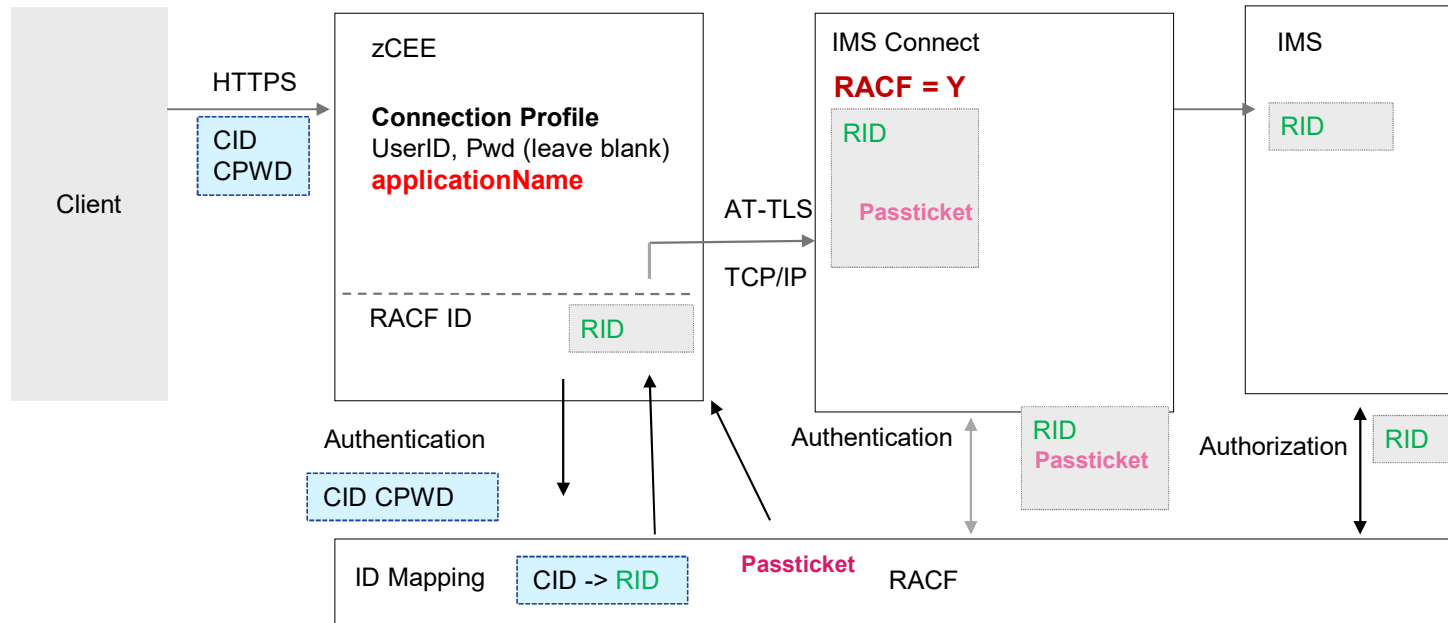
Or...
For environments that need the end-user userid to be passed into IMS....

Is implementing a modified HWSJAVA0 exit routine in IMS Connect acceptable?

or.. Do you have IMS Connect Extensions?

Scenario 4 – Client Identity + Passticket + RACF=Y

FID FPWD = User ID, Password are NOT specified in the connection profile
but the connection profile specified applicationName for generating passticket
CID CPWD = User ID, Password of the originating client (Basic Auth or Cert, etc)
RID = RACF Mapped ID + Passticket



Passtickets (in lieu of passwords)
Are generated on the zCEE side by the IMS SP and authenticated on the IMS connect side

zCEE V3.0.33
Supports both IMS
Service Provider
and IMS DB
Service Provider

To review: the user ID is determined based on the following rules:

- If a user name and a password are specified in the connection profile, the user name and the password are used for SAF authentication with IMS Connect.
 - Note: this will be 'generic' to all the requests using this profile
- If the end-user id is to be used (make sure that the connection profile does not contain a user name/password) :
 - The user ID is extracted from the request subject from z/OS Connect EE (no passwords)
 - ZCEE must be configured with SAF registry authentication and the subject must be 8 bytes or less in order for the IMS service provider to retrieve the user ID from the request subject.
 - > if passticket support is provided, the passticket is generated and passed to IMS Connect and used for authentication
 - > If passticket support is not available, then if defined, the 'generic' IMS technical password is the SAF password for the user – otherwise, no password is provided but if RACF=Y in IMS Connect then trusted user support can be enabled with HWSJAVA0
 - If the request subject is more than 8 bytes, or authentication is disabled on z/OS Connect EE, the IMS service provider retrieves the user ID from the technical ID, an IMS mobile global element that is specified during initial installation and setup. The IMS technical password is the SAF password for the user.
 - If the technical ID is left blank, the IMS service provider uses the z/OS Connect EE started job user ID. The IMS technical password, if specified, is the SAF password for the user.
- The IMS technical groupname, if specified, is the SAF groupname.

And if you have Tooling

IMS Connect Extensions V 3.1 - Workload Rules for IP addresses

Override User id

Override the user ID defined in the request message with the specified user ID. Workloads will be processed using the specified user ID, regardless of what is contained in the request.

Only override blank userids

When this option is selected, the user ID specified in the Override User id field is only used when no user ID is supplied in the request message.

Trust user id

Treat all messages from this IP address or IP address range as trusted users in IMS Connect. SAF password verification in IMS Connect is not performed.

```
EDIT                                     IP Address Rule
Command ==> _____

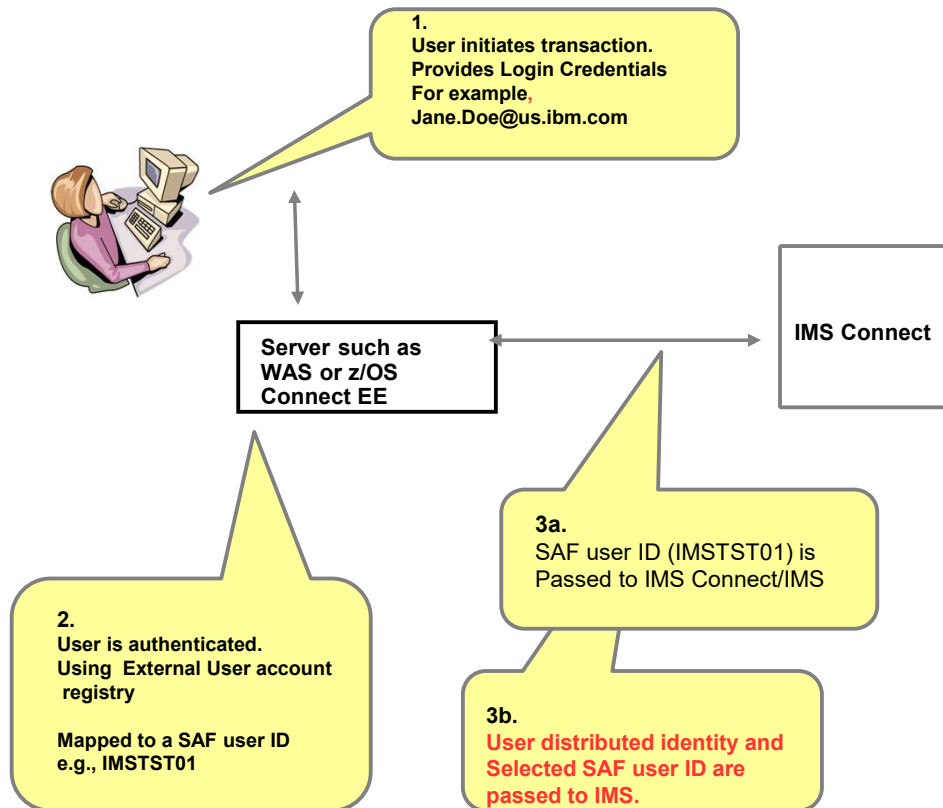
Name . . . . : SRVR001
Description . . Rule for SRVR001 _____

Apply rule to:
1 1. System . . . . . HWSOPGS1 +
2. Group . . . . . _____ +
3. All systems

Condition:
IP Address . . . . . 193.23.88.122 _____

Options:
/ Override User id VIP001 _____
_ Only override blank userids
_ Trust user id
< _____
```

Distributed Credential Mapping (New in IMS 15) and optional Propagation



- **IMS 15 provides support for propagating the distributed (network) credential for purposes of auditing and logging**
 - IMS Log records
 - > Up to 246 bytes for client end-user identity **Plus** up to 254 bytes for a realm or registry identity
 - RACF SMF x'80' record
 - > Up to 255 bytes
 - IMS security exits
 - IMS application program with the INQY call
 - IMS application ICAL for outbound requests
- Supported clients
 - IMS 15, IMS Connect 15,
 - IMS TM Resource Adapter V15 (IMS TMRA) with
 - WAS
 - Liberty and z/OS Connect with the IMS Service provider
 - V15 OTMA C/I
 - IMS Connect V15 RYO clients
- **IMS Connect and IMS rely on the SAF userid ID for security**

How Simple or Complex is the IMS transaction interaction?

Ensure that someone who knows the architecture of the application is involved in the process



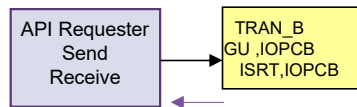
Otherwise you can run into problems that could/should have been avoided

Simple

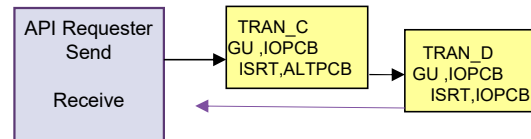
- Considerations – The IMS Service Provider uses CM1 Send-Commit (tpipe is the port number)
Can use CM0 Commit-send

- **Simple interactions**

- Straightforward API implementation using the API Toolkit



Input and output layout for the
Tool are from TRAN_B



Input layout for the tool is for TRAN_C
And output layout is from TRAN_D

- **AND:**

- *IMS transaction is non-conversational*
- Cobol/PLI programs provide the input/output message layouts (copybooks or include files)
- Any program-to-program switches are done with
 - Non-express ALTPCBs
 - Final switched-to transaction responds to the IOPCB
 - No spawning of multiple transactions – switching is simply from one transaction to another

Complex – Example 1

— What could cause complexity in this environment?

- No available copybooks to provide the toolkit to build the services?

Analysis will need to be done of the input/output IMS messages to create the layouts

- What about having to determine the flow of the application business logic when multiple transactions are involved in program-program switches?

The exposure to REST API implementation means that the input layout needs to be tied to the final output to determine which output message from a switched-to transaction matches the original input message

- > IMS application architects should be involved to determine this flow

— Possible resolutions:

- Execute the transaction flow and analyze

- 01/03 log records

Run DFSERA10 / DFSERA30 and assemble the IMS Log record DSECTs

- Or: take a trace and analyze the trace records
- Or use tooling

Log Record Analysis

1. Run DFSERA10 / DFSERA30

```
OPTION PRINT OFFSET=5,FLDTYP=X,VALUE=01,FLDLLEN=1,COND=E,EXITR=DFSERA30
OPTION PRINT OFFSET=5,FLDTYP=X,VALUE=03,FLDLLEN=1,COND=E,EXITR=DFSERA30
```

To get:

-03 RECORD -

```
00000000 000000 022F0000 03C38210 040000D1 040000D1 01DE0000 C9C9D4E2 40404040 D4C64F35 *....CB....J....IIMS MF3*
00000020 000020 C9A40C53 C9C9D4E2 40404040 D4C64F35 C9A40C53 00000000 00000000 00000000 *IU..IIMS MF3IU.....*
00000040 000040 00408100 C8000000 00000000 00000000 00000000 00000000 00000001 *.A.H.....*
```

2. Assemble the IMS log record DSECTs to get the record layout

ILOGREC ID=

110+MSGLRZZ DS H	ZZ FIELD RESERVED FOR QSAM
111+MSGLCODE DS X	LOG CODE (01 OR 03)
113+MSGFLAGS DS X	MESSAGE FLAGS
114+MSGFFRST EQU X'80'	FIRST RECORD OF MSG, FULL PREFIX
115+MSGFLAST EQU X'40'	LAST DRRN (RECORD) OF MESSAGE
116+MSGFCANC EQU X'20'	MESSAGE CANCELED
117+MSGFNRRQ EQU X'10'	NON-RECOVERABLE QUERY MESSAGE
118+MSGFNRR EQU X'0F'	L/O NIBBLE HAS LOGICAL QUEUE NO....
...	

3. Use the DSECT layout to analyze the log records

OR:

Use a tool like IMS Problem Investigator to access and analyze the IMS log records in a more user-friendly fashion

1 Select a record to view all of its fields

Forwards / Backwards . . 00.00.00.000100 Time of Day . . 01.10.30.000000
Code Description Date 2017-10-27 Friday Time (Local)

01 Input Message UTC=17.10.56.568088 TranCode=ATMWDRAW Userid=NEWYORK LTerm=NEWYORK
Terminal=NYATM001 OrgUOWID=I9DE/BE8300F4C92D4A23

08 Application Start UTC=17.10.56.574100 TranCode=ATMWDRAW Region=0002
RecToken=I9DF/0000000300000000 RegType=MPP TClass=01 TPrty=08

31 DLI Form ==> + Top of data Format ==> FORM
UTC=17.10.56.568088
OrgUOWID=I9DE/BE8300F4C92D4A23

5616 Star RegType=MPP TClass=01 TPrty=08

03 Outp UTC=17.10.56.574100 TranCode=ATMWDRAW Region=0002
RecToken=I9DF/0000000300000000 RegType=MPP TClass=01 TPrty=08

Field Zoom

+0007 MSGFPPADL... 94 Prefix Additional Info Flag

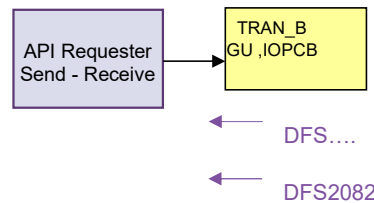
On MSGFPRSP... 80 Response Mode
Off MSGACMD... 40 Scheduled APPL issued 'CMD'
Off MSGAOIUE... 20 Message generated by AOI user exit
On MSGSYSEG... 10 System Segment exists
Off MSGSSPND... 08 Message is on SMB Suspend queue
On MSGFPINR... 04 Input message is non-recoverable

MSGDRBN... 00000000
tem ID = 81
MSGCFLG1... 00
MSGCQSF1... 00

2 Zoom on a field to view a detailed description of its value

Complex – Example 2

- If there are problems in the IMS environment, or the IMS transaction abends, or the transaction does not reply to the IOPCB



- The REST API may receive an unexpected response
 - If the target transaction does not ISRT to the IOPCB or ALTPCB
The API requester will get a DFS2082
 - Or, If the target transaction experiences an error/problem on the IMS side
The API requester will get a DFS... message
- Solution
 - Design the REST API to expect either the output response from the transaction or an IMS DFSxxxx message

Complex – Example 3

— The IMS transaction environment has application security or needs an LTERMname

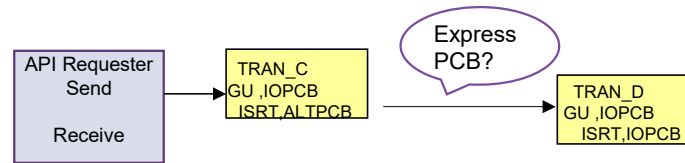


- LTERMnames may be required by the application(based on older 3270 environments)
 - Determine if a generic LTERMname can be used
 - *Define it in the connection profile in z/OS Connect*
 - Otherwise, some possible solutions:
 - Define the value needed for LTERMname in the REST API input message (e.g., LLZZ trancode data **overridename**)
 - > *Specify a dummy value in the LTERMname value of the connection profile*
 - > *Write an IMS OTMA exit (DFSYIOE) or IMS Connect exit (HWSJAVA0) to recognize the dummy value from the connection profile and:*
 - *Remove the overridename from the input message*
 - *Use the value to pass it in as an LTERMname*

Other possibilities would depend on the specific customer environment

Complex – Example 4

- The IMS transaction program-program switch uses an Express PCB



- The REST API request will hang
- Solution:
 - > Change the Express PCB to Non-express PCB



- BUT... the real issue on a hang is troubleshooting... so

Complex – Example 4 ...

— The IMS transaction program-program switch uses an Express PCB ...

- **Troubleshooting:**

- IMS Connect

Recorder trace can determine if any problems exist in IMS Connect

> Requires the trace to be turned on while the interaction occurs

Easier alternative would be to use a tool like IMS Connect Extensions

- IMS log records – show the flow in IMS that can cause the hang

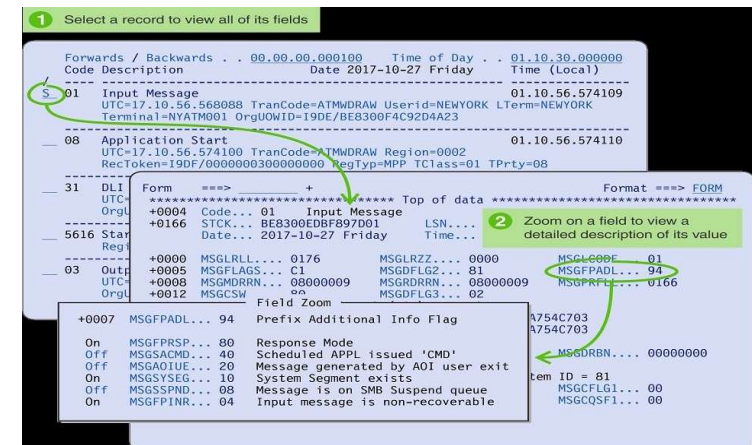
Or use tooling. e.g, IMS PI to drill down through the log records

Manual process

1. Run DFSERA10/DFSERA30 to extract IMS log records
2. Using log records DSECTs, determine offset for MSGIDSTN and MSGODSTN

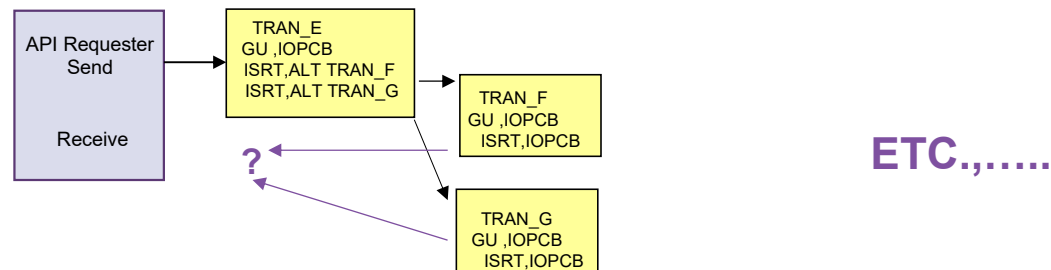
	MSGIDSTN	MSG output Destination
01 (CM1)	FDFFFFFF 30738760	E3D9F0F1 40404040 (target is the TR01 txn)
pgm-pgm switch		
03 (CM1)	FDFFFFFF 30738760	E3D9F0F2 40404040 (target is the TR02 txn)
Shows problem is that the interaction is now asynchronous (CM0)		
pgm-pgm switch		
03 ----->	FEFFFFFF 305AC948	E3D9F3F3 40404040 (target is the TR33 txn)
03 (CM0)	FEFFFFFF 305AC948	FEFFFFFF 305AC948 outbound
hang		

MSGIDSTN switches from CM1 synchronous - FDFFFFFF – (TIB) to CM0 (asynchronous - FEFFFFFF – (QAB)



Complex – Example 5

- The target transaction spawns multiple transactions - which should respond back to the API requester?

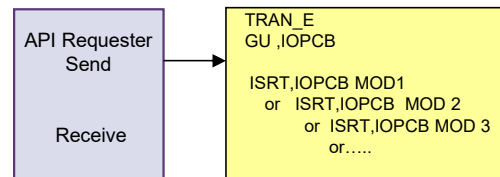


- If the target transaction spawns multiple transactions, only one IOPCB response can be sent to the API requester
- Solution
 - **Determine which transaction IOPCB message should reply to the request**
 - > **OTMAASY = Y | N | S** parameter
 - Controls which transactions is switched to synchronously (eligible to reply to the request)
 - Y** – First RESPONSE program scheduled after the switching transaction ends is scheduled synchronously
 - N** – First program (RESPONSE or NONRESPONSE) scheduled after the switching tran ends is scheduled synchronously
 - S** – The first message ISRTed to a non-Express ALTPCB will be scheduled synchronously

➤ **Note that the other IOPCB message(s) will be purged**

Complex – Example 6

- The target transaction replies using one of many possible output layouts



- Existing 3270-based transactions may specify extended attributes in the program
Extended attributes are 2-bytes per message field, e.g., color, highlight, etc. that affect the output message layout
- Note that if the program specifies these attributes, they will be included as part of the application output message and will require the API requester to recognize them as such
- Possible actions
 - Review the architecture with program architects
Use tooling such as IBM ADDI to analyze the program
 - Determine the possible paths/layouts that can be used
 - Explore the ability to create a copybook for the API toolkit which uses Redefines
 - Determine if a more generic output message layout to include all the attributes can be used

Overall Recommendations

- On the IMS side, ensure that the application architecture is understood
 - Determine if the application is ‘simple’
 - Anticipate complexity or potential show-stoppers
 - Determine if it would be easier to write/modify the IMS applications

Possibility: Orchestration transaction that uses IMS synchronous Program Switch support to invoke existing transactions

- For problems,
 - Log records, traces, or tools

Tooling

IMS Connect / IMS / Application

Journaling and control: IMS Connect Extensions (IMS CEX)

Problem determination: IMS Problem Investigator (IMS PI)

Performance: IMS Performance Analyzer (IMS PA)

Omegamon for IMS

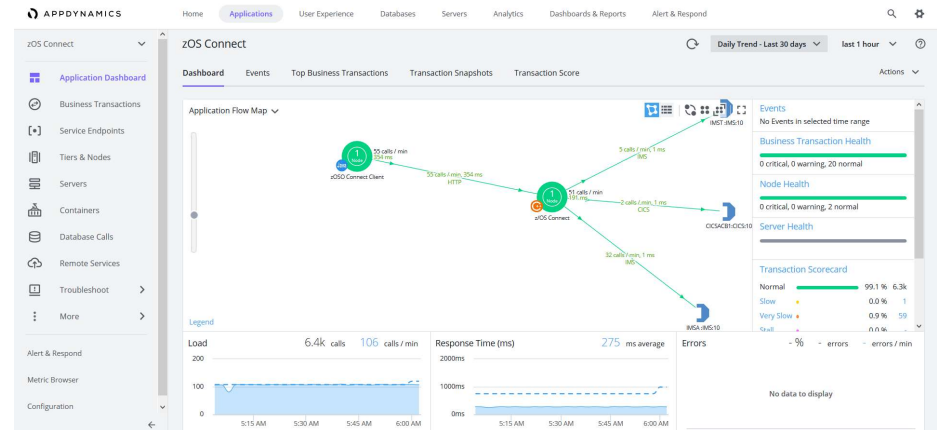
Application workflow: ADDI

z/OS Connect

Omegamon for JVM – can monitor z/OS Connect API utilization

End-to-End tracking

IBM z APM Connect - Can be used to determine/ target problem areas



Overall Recommendations ...

— For problem determination, turn on tracing in z/OS Connect server.xml file

- `<logging maxFileSize="20" maxFiles="10" traceFileName="imsmobile.log" traceFormat="BASIC" traceSpecification="com.ibm.ims.gateway*=FINEST:com.ibm.ims.zconnect.provider*=FINEST"/>`

The screenshot displays the z/OS Connect Enterprise Edition interface. On the left, the Project Explorer shows the file structure, including the `imsmobile.log` file. The central pane shows the `server.xml` configuration file with the logging section highlighted. The right pane shows the trace output from `imsmobile_19.09.19_01.26.58.0.log`, which includes details about the security distinguished name, realm name, and network security credentials. The trace output also shows a detailed log of the RECEIVE BUFFER operation, including the ASCII and EBCDIC representations of the data.

workspace - z/OS Connect Enterprise Edition - RemoteSystemsTemp...

File Edit Navigate Search Project Run Design Window Help

Project Explorer

z/OS Connect EE Servers Remote Systems

ffdc

state

imsmobile.log

imsmobile_19.09.19_01.26.58.0.log

imsmobile_19.09.25_11.17.05.0.log

imsmobile_19.10.08_13.35.51.0.log

imsmobile_19.11.19_13.15.49.0.log

imsmobile_20.01.30_18.23.09.0.log

imsmobile_20.01.30_18.47.16.0.log

Connection: zserver...S.CENTERS.IHOST.COM

server.xml

<!-- z/OS Co
RACF us
-->

<include locatio
<include locatio
<include locatio

Design Source

z/OS Connect Enterprise Edition - RemoteSystemsTempFiles/ZSERVEROS.CENTERS.IHOST.COM/var/zosconnect/servers/suziew/lo...

gate Search Project Run Window Help

server.xml imsmobile_19.09.19_01.26.58.0.log IMS Explorer Task Launcher

Security Distinguished Name: null

Realm Name: null

Propagate network security credentials: true

VTNO

[8/7/19 20:40:10:585 GMT] 00011bbb IMSGatewaySer 3 trackingToken: C2C1D8010018C4C5D4D6D7D3E740C5E2E8E

[8/7/19 20:40:13:164 GMT] 00011bbb services 3 [72635]

[ibm][ims][tran] RECEIVE BUFFER: (ASCII) (EBCDIC)

[ibm][ims][tran] 0 1 2 3 4 5 6 7 8 9 A B C D E F 0123456789ABCDEF 0123456789ABCDEF

[ibm][ims][tran] 0000 005D0300C5D5E3D9 E840E6C1E240C4C9 .].....@...@..)..ENTRY WAS DI

[ibm][ims][tran] 0010 E2D7D3C1E8C5C440 4040404040404040@..... SPLAYED

[ibm][ims][tran] 0020 4040404040404040 40404040C4C9E2D7 @.....@..... DISP

[ibm][ims][tran] 0030 D3C1E840D3C1E2E3 F14040404040C6C9 ...@.....@..... LAY LAST1 FI

[ibm][ims][tran] 0040 D9E2E3F140404040 F860F1F1F160F1F1@..... RST1 8-111-11

[ibm][ims][tran] 0050 F1F1C4F0F161D9F0 F1F0F0F0F1a..... 11D01/R010001

[8/7/19 20:40:13:164 GMT] 00011bbb services 2 [72635] IMSGatewayServiceImpl: Exiting : executeTr

Host Connections Properties Progress Problems

Connections Credentials

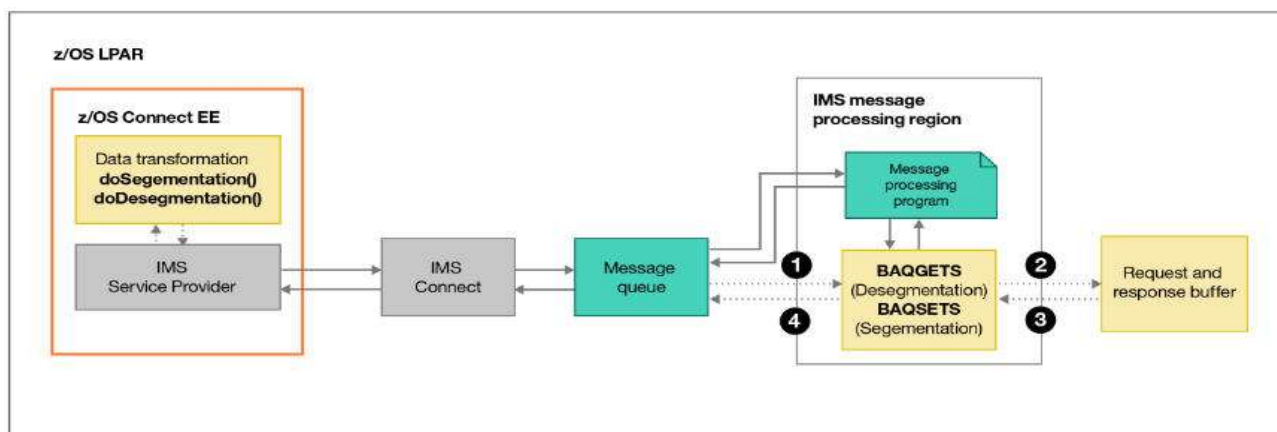
Writable Insert 4441:45

And Even More Capabilities for APIs

Support for Large Data Structures

— zOS Connect EE V3

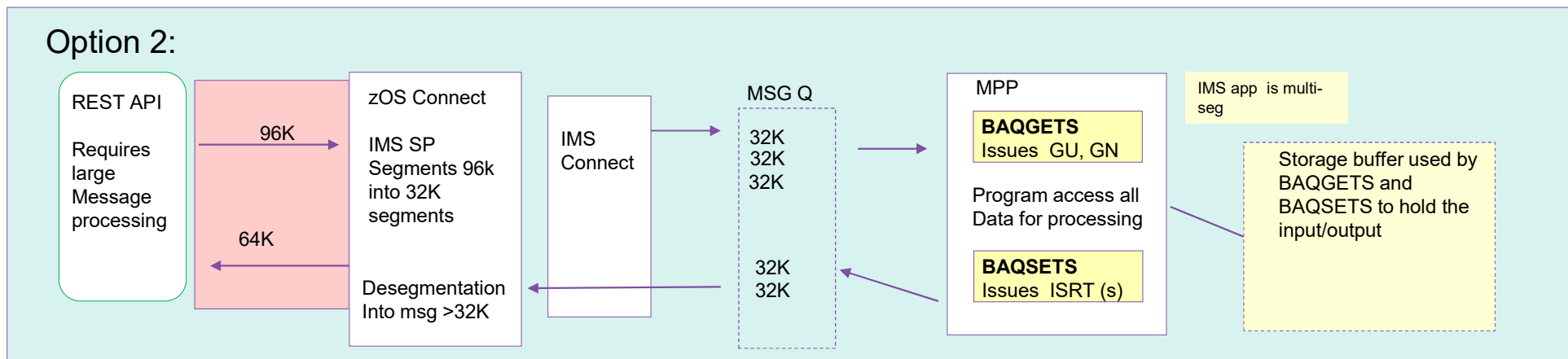
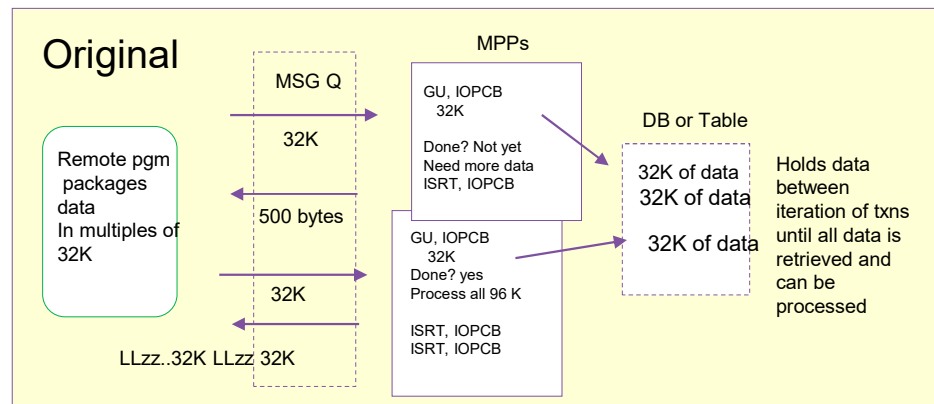
- Supports large data structures (>32K IMS standard segmentation)
 - IMS applications must invoke the supplied sample COBOL utilities BAQGETS and BAQSETS instead of interacting with the IMS message queue directly
 - Sample segmentation utilities that are bound with the Cobol application (PL/I later)



Large IMS data structure support is automatically enabled when the IMS service provider detects that an IMS service, created by using z/OS Connect EE API toolkit, is based on an input or output transaction message that consists of a single segment whose data structure has a maximum length that exceeds 32 KB.

Support for Large Data Structures ...

— Examples of a use case



Large Data Structures - Recommendations

— Review the IMS application environment to determine the value proposition

- If the remote program sending the API request simply need to sends a large unsegmented message (larger than 32K) or retrieve a large unsegmented message
 - If the IMS application is already multi-segment, then the use of this support could allow the IMS application program to remain unchanged

— Ensure the IMS application is multi-segment

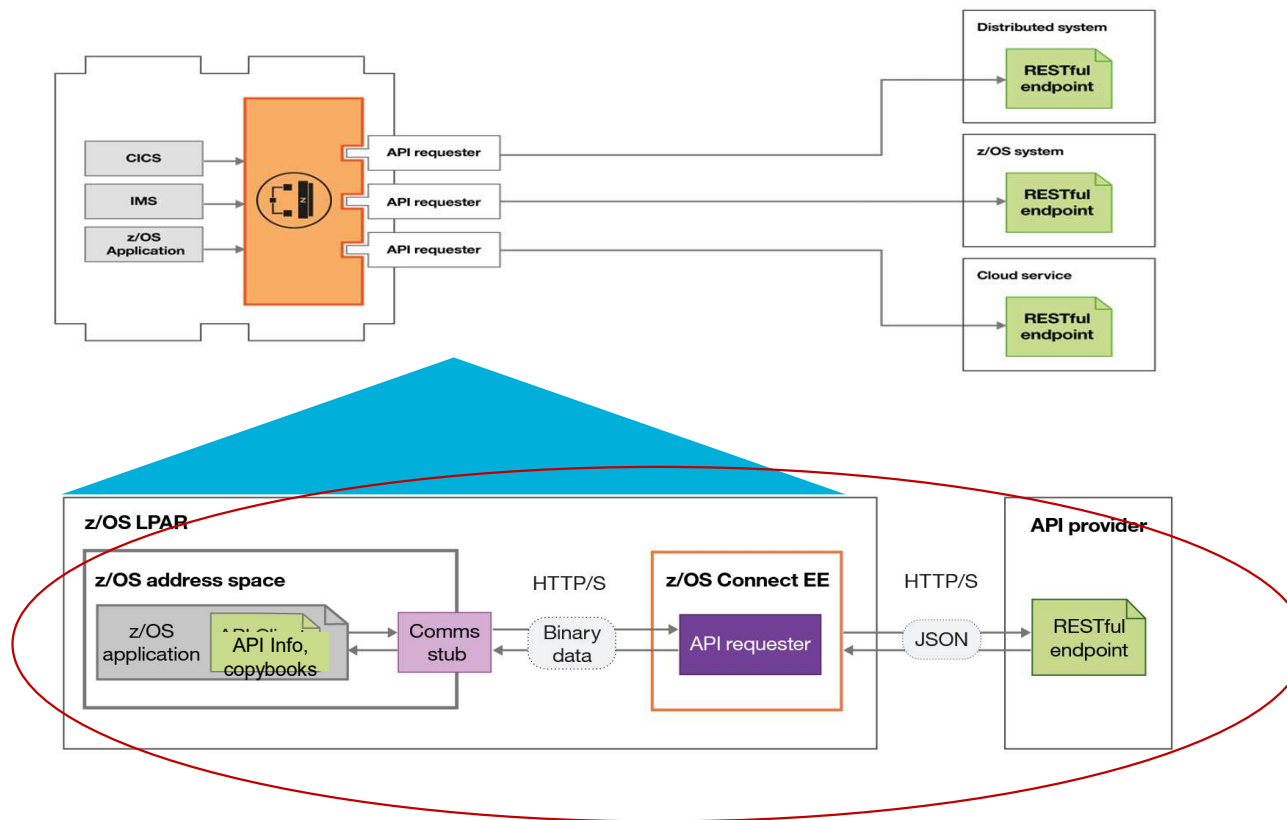
- E.g., Modify the program if it is currently single segment
 - IMS code to do GU, GN calls
 - Or, replace the IMS message calls with the BAQGETS/BAQSETS'

— Understand the impact of BAQGETS/BAQSETS

- **Only use these if:** the 'large' message is >32K
- Large data structures sent in through the API must not begin with an LLZZ or LLZZ<TRANCODE> prefix
 - The BAQGETS utility generates segment prefixes with the first segment prefix set to the transaction code that is specified in the API toolkit
- PH09920 – adds a 'space delimited' option to the API toolkit v3.0.6.5 or v 3.2.6.5 and z/OS Connect EE V 3.0.21
 - option to place only a single blank space character after the trancode, preserving any user data bytes at the end of the 8 byte trancode area.

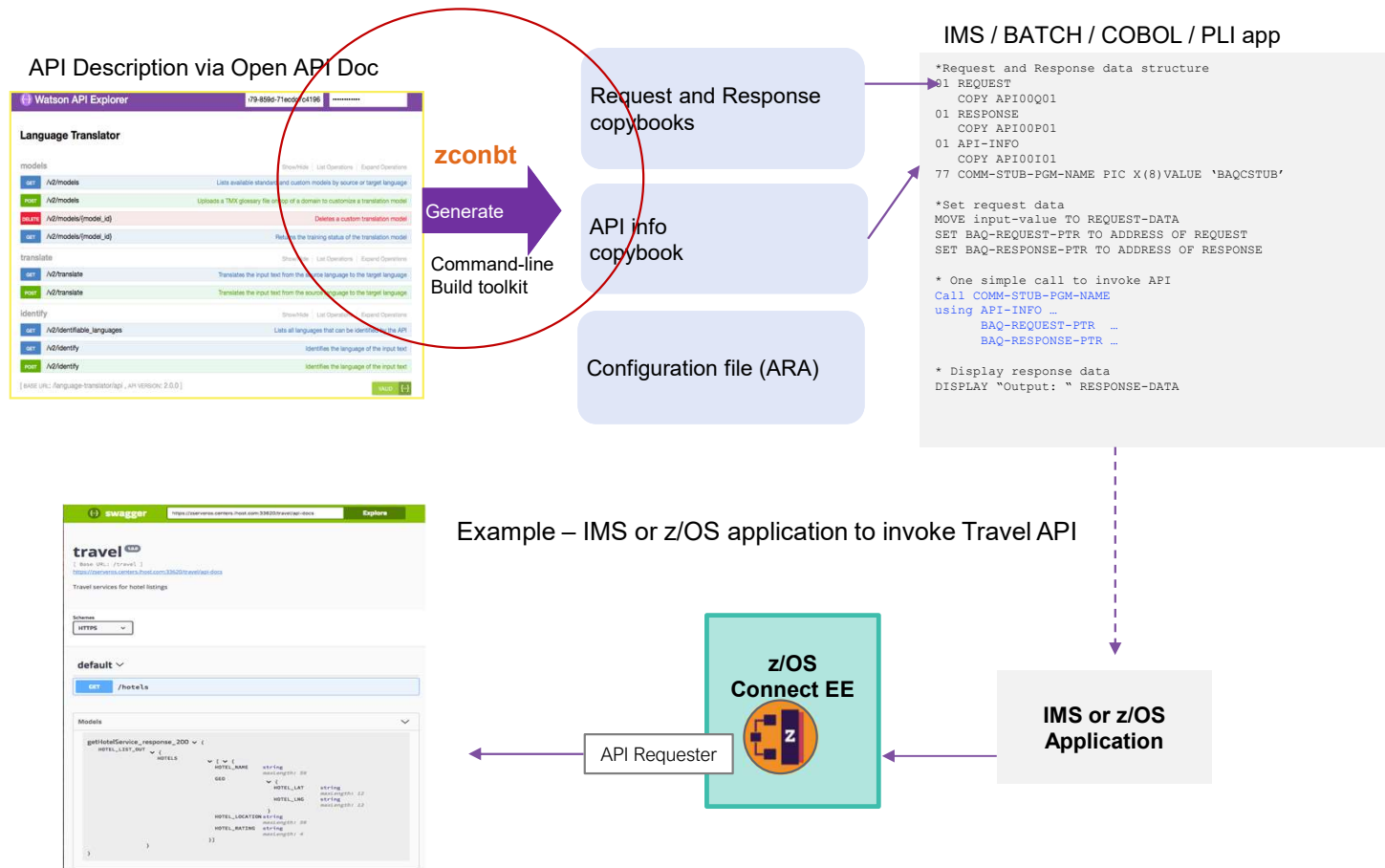
And the Ability to Call out: IMS applications → REST APIs

- z/OS Connect EE V3 allows IMS applications to be the API requester



API requester code generation for z applications

z developer does not need to write complex code to parse JSON messages nor to handle HTTP calls



IMS Compile link-edit RENT with Cobol V4 +

```
//COMPLNK1 EXEC IGYWCL,
//  LNGPRFX='IGYV5R20',
//COBOL.SYSIN      DD DISP=SHR,
//  DSN=IMS.APPL.SOURCE.COBOL(HOTELAPB)
//COBOL.SYSLIN     DD DISP=SHR,
//  DSN=IMS.APPL.COBOL.OBJECT(HOTELAPB)
//LKED.OBJECT      DD DISP=SHR,
//  DSN=IMS.APPL.COBOL.OBJECT
//LKED.RESLIB      DD DISP=SHR,
//  DSN=IMS.SDFSRESL
//  DD DSN=IMS.APPL.COBOL.LOAD,DISP=SHR
//  DD DSN=ZOSCON.V3R0.SBAQLIB,DISP=SHR
//LKED.SYSLIB      DD DSN=SYS1.CSSLIB,DISP=SHR
//  DD DSN=SYS1.SCEELKED,DISP=SHR
//  DD DSN=SYS1.SCEELKEX,DISP=SHR
//LKED.SYSLIN      DD *
//      INCLUDE OBJECT(IMMRAP10)
//      INCLUDE RESLIB(BAQCSTUB)
//      INCLUDE RESLIB(DFSLLI000)
//      ENTRY IMMAPI
//      NAME IMMRAP10(R)
/*
//LKED.SYSLMOD      DD  DSN=IMS.PGMLIB.PDS(HOTELAPB),DISP=SHR
//      (or PDSE if Cobol 5.1+)
/*
```

Name	Prompt	Alias-of	Size	TTR	AC	AM	RM
BAQCSTUB		BAQWEBT	000007EC	000004	00	31	ANY
BAQWEBT			000007EC	000004	00	31	ANY

Configure the IMS Dependent Region JCL

- CEEOPTS specify LOCALHOST or z/OS Connect URI and port number

```
//CEEOPTS DD DSN=IMS.PROCLIB(MPRLEOPT),DISP=SHR
```

or

```
//CEEOPTS DD *
```

```
POSIX(ON),
```

```
ENVAR("BAQURI=HTTP://ZOSSEE.SERVER.IBM.COM","BAQPORT=7777",
"BAQVERBOSE=ON","BAQTIMEOUT=60")
```

- STEPLIB

```
//STEPLIB DD DSN=IMS.SDFSRESL,DISP=SHR
```

```
//          DD DSN=IMS.PGMLIB,DISP=SHR
```

```
//          DD DSN=IMS.PGMLIB.PDS,DISP=SHR
```

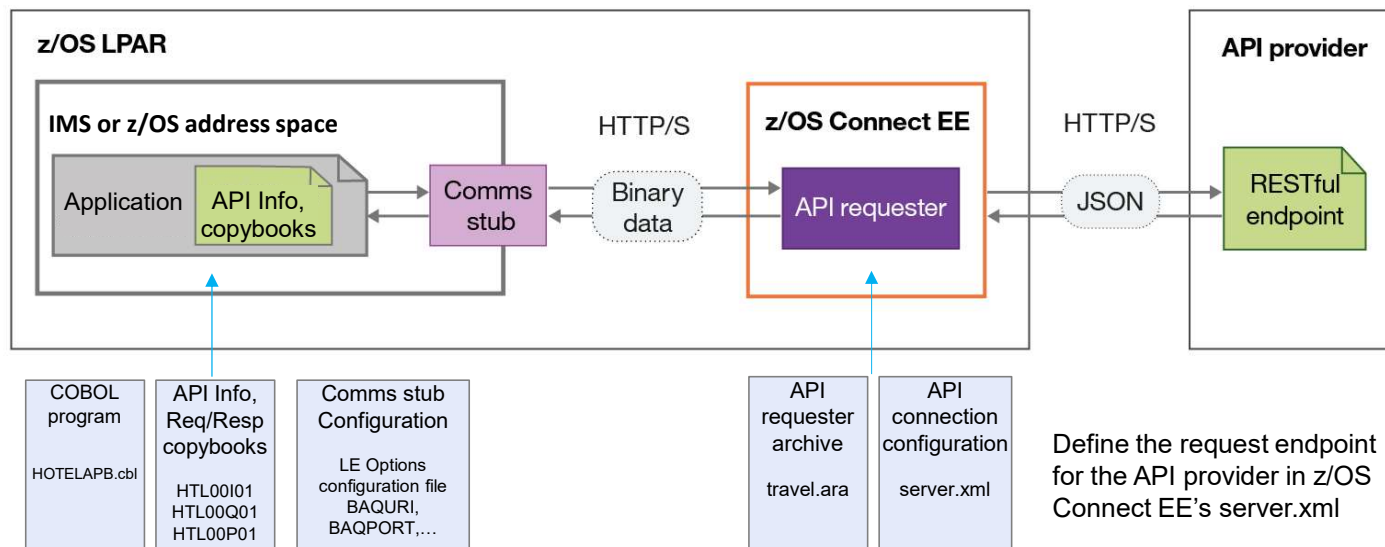
```
//** if COBOL 5.1+ DD DSN=IMS.PGMLIB.PDSE
```

```
//          DD DSN=ZOSCON.V3R0.SBAQLIB,DISP=SHR
```

```
//*          BAQCSTUB load module
```

Deploy and Configure the application

Deploy and Configure API requester in z/OS Connect EE



z/OS Connect EE V3.0.5.0

PI92254 provides a sample IMS app for calling out using REST APIs in <zceehlq>SBAQSAMP

```
<!-- apiRequester definition -->
<zconnect_apiRequesters updateTrigger="polled"/>
<zconnect_authData id="apicAuthData"
  user="userID"
  password="passwd"/>
<zconnect_endpointConnection id="hotelConnRef"
  host="https://api.us.apiconnect.ibmcloud.com"
  port="443"
  sslCertsRef="defaultSSLConfig"
  basicAuthRef="apicAuthData" />
```

API Requester Recommendations

- Web Enablement Toolkit
 - Implement latest maintenance
 - E.g., OA54670(UA96376) handles communication interruptions
- Preload the program that issues API Requester callout
- LE Cobol runtime option RTEREUS should not be used in POSIX environment
 - Can cause issues
- Set TLIM = 1
 - If there any abends the MPR is automatically terminated
 - If TLIM is >1, pseudo abends cause DSFRRC00 to reattach DFSPCC20 and problems can occur
- **Make Sure you are up to date on maintenance**

API Requester Recommendations...

— As with other IMS applications that synchronously call out from the program, the interaction holds IMS resources:

- Consider creating a class and specific dependent regions for this program
 - Fences this program from impacting other transactions that could be classed to run in the same region
 - Facilitates problem determination if the program stays hung
- Any database calls that were issued prior to the API request will hold locks
 - If the IMS database call was an update, then this could have performance implications for other transactions
 - If the IMS database call was a read, consider using the RLSE call which can release IMS DB read locks
 - Provides the ability to release IMS DB read locks
 - Update locks are not released

Fast Path DBs - releases all locks held for unmodified data

Full Function DBs - release locks held by the DB PCB that is referenced

Note: after the RLSE call, all database position information is lost so the application would have to reposition

- Example:
 - *Program reads IMS DBs*
 - *Prepares message for callout with a wait (ICAL, MQPUT_MQGET...) which will hold the region occupied and hold locks*
 - *RLSE releases the read locks and positions on database*
 - *Keep information on positioning if needed*
 - *After the callout message is returned, re-position on databases if needed and continue processing*

API Requester Recommendations ...

- Ensure that during **testing**, BAQVERBOSE is turned on in the dependent region
 - Point to the LE options member from the CEEOPTS DD
 - Provides information about unusual abends (e.g., S01F) or hangs which can occur

The left screenshot shows a terminal window with the following command:

```
EDIT IMS. IMSA.PROCLIB(MPRLEOPT) - 01.05 Columns 00001 00072
***** Top of Data *****
000001  POSIX(ON)
000002  ENVAR("BAQVERBOSE=ALL")
000003  "BAQURI=ZSERVERS.CENTERS.IHOST.COM","BAQPORT=23628")
***** Bottom of Data *****
```

The right screenshot shows the SDSF output of the IMS job, displaying verbose output for the Web Toolkit connection. The output includes the following lines:

```
SDSF OUTPUT DISPLAY IMSAMSG3 JOB03583 DSID 103 LINE 40 COLUMNS 02- 81
COMMAND INPUT ==>
20-02-13 18:53:20:890120 (setConn) -> (0x23206420, ZSERVERS.CENTERS.IHOST.COM)
20-02-13 18:53:20:890130 (setConn) -- Web Toolkit verbose output enabled
20-02-13 18:53:20:890133 (HSet) -> ()
20-02-13 18:53:20:890135 (HSet) -- Calling hwthset (0x2322EBE8, 0x2322EB40, 0x2322EB40)
20-02-13 18:53:20:890150 (HSet) -- Return from hwthset {0}
20-02-13 18:53:20:890153 (HSet) <- {0}
20-02-13 18:53:20:890156 (setConn) == pm_hostname=ZSERVERS.CENTERS.IHOST.COM
20-02-13 18:53:20:890158 (HSet) -> ()
20-02-13 18:53:20:890161 (HSet) -- Calling hwthset (0x2322EBE8, 0x2322EB40, 0x2322EB40)
20-02-13 18:53:20:890182 (HSet) -- Return from hwthset {0}
20-02-13 18:53:20:890184 (HSet) <- {0}
20-02-13 18:53:20:890188 (setConn) == pm_port=23628
20-02-13 18:53:20:890190 (HSet) -> ()
20-02-13 18:53:20:890192 (HSet) -- Calling hwthset (0x2322EBE8, 0x2322EB40, 0x2322EB40)
20-02-13 18:53:20:890198 (HSet) -- Return from hwthset {0}
20-02-13 18:53:20:890200 (HSet) <- {0}
20-02-13 18:53:20:890202 (setConn) == pm_timeout=Receive timeout 10
20-02-13 18:53:20:890230 (HSet) -> ()
```

At the bottom of the right window, the status bar indicates: "Connected through TLS1.2 to secure remote server/host zservers.centers.ihost.com using 128".

Putting it Together... What About Performance?

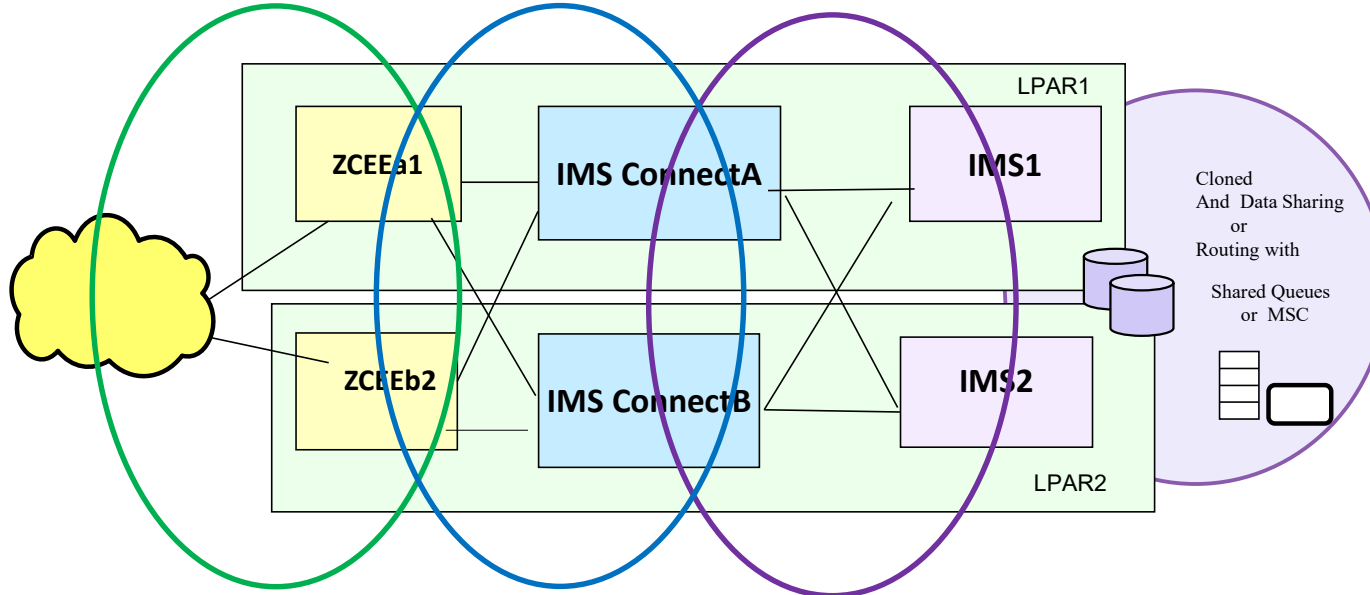
— Performance Paper – z/OS Connect Enterprise Edition V3 Performance Evaluation Using IMS

- <https://www.ibm.com/downloads/cas/49XVRX2O>
- Conclusion: Performance scalability in terms of transaction rate and CPU efficiency as concurrent users increase
- Tests were conducted in a controlled environment using message-only transactions (No DB)
 - NOTE: While performance in specific production environments will vary, results of IBM internal testing in a controlled laboratory environment revealed that the z/OS Connect EE V3 is capable of performing as described below:
 - **API Provider Workload Performance:**
 - Peak transaction rate of 21,000 transactions per second with 1KB message sizes
 - Peak transaction rate of 8,900 transactions per second with 8KB message sizes
 - Peak transaction rate of 3,500 transactions per second with 30KB message sizes
 - **API Requester Workload Performance:**
 - IMS application invoking a RESTful API -- peak transaction rate of 5,100 transactions per second with 1KB message sizes
 - IMS application invoking a RESTful API -- peak transaction rate of 3,800 transactions per second with 8KB message sizes
 - IMS application invoking a RESTful API -- peak transaction rate of 2,100 transactions per second with 30KB message sizes
 -

But.. As the environment grows
So does the need for high availability and resiliency

And... With Expanding Environments

- z/OS Connect – IMS Connect - IMS
 - An increasing need to configure for High Availability and Resiliency
 - **Redundancy, Failover and workload balancing**

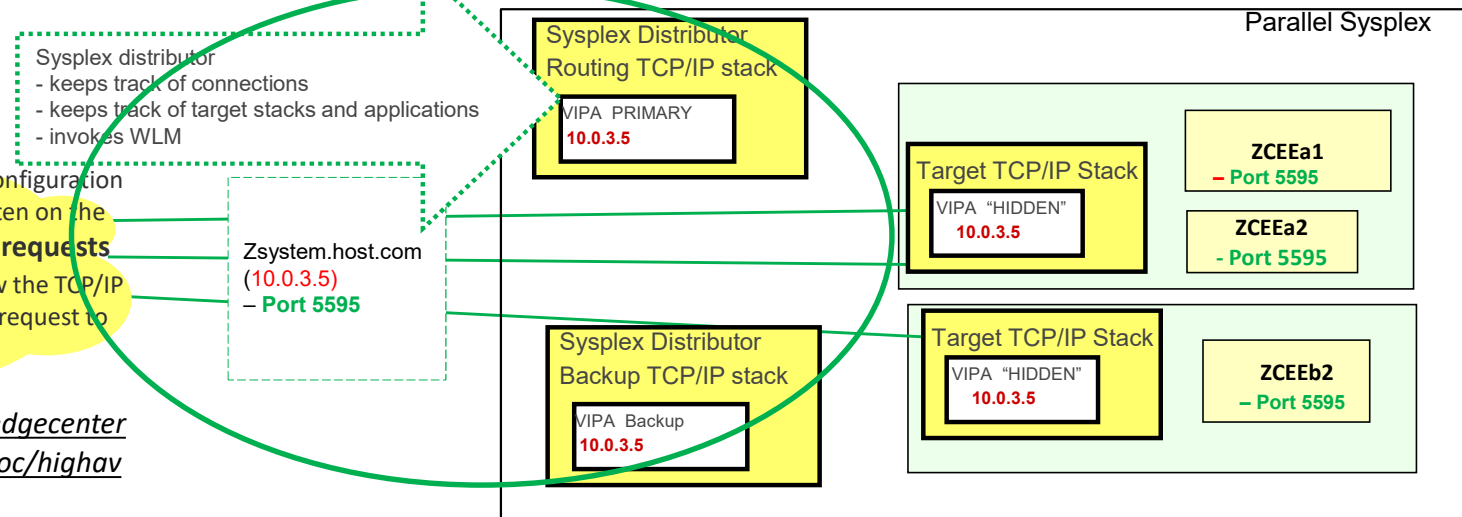


And... With Expanding Environments

- A **z/OS Connect EE High Availability Environment** can leverage TCP/IP load balancing technologies to distribute connections from clients across multiple servers
 - Both for requests to invoke APIs and services in z/OS Connect EE servers, or RESTful API requests from z/OS applications
 - **Sysplex Distributor** - distributes requests across multiple ZCEE servers in a Parallel Sysplex
 - Dynamic VIPA and Automatic VIPA takeover
 - **TCP/IP port sharing** - load balances requests across multiple z/OS Connect EE servers in **the same** z/OS LPAR.
 - **WLM classification** - classifies HTTP traffic in a ZCEE server so that it can be managed by z/OS workload management (WLM) along with other work in the system.

By configuring the httpEndpoint in the server configuration file of multiple z/OS Connect EE servers to listen on the same shared TCP/IP port, **clients can send requests to a single hostname and port** and allow the TCP/IP load balancing technologies to distribute the request to one of the available servers

https://www.ibm.com/support/knowledgecenter/SS4SVW_3.0.0/com.ibm.zosconnect.doc/highavailability/ha_concepts.html



And... With Expanding Environments

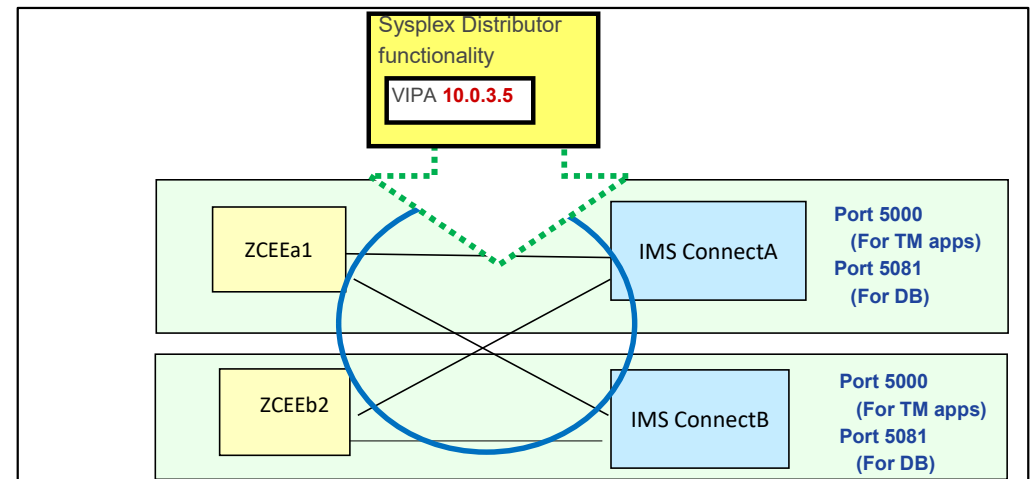
— z/OS Connect – IMS Connect

- Since IMS Connect is the TCP/IP socket server for IMS, **the same load balancing technologies** can be used to access the IMS Connect (hostname and port number)
 - **Sysplex Distributor, TCP/IP port sharing, WLM**

Connection profiles in ZCEE define the target IMS Connect and port number:

```
Example: IMS_CF1 ...  
<properties.gmoa hostName="zsystem.host.com"  
                                     or "10.0.3.5"  
      portNumber="5000">  
.....
```

- **Instead** of full workload balancing between the target IMS Connect systems
 - If the design is to always go to the IMS Connect on the LPAR that is local to ZCEE and only route requests to the IMS Connect on the other LPAR
 - *Define a common listener port that is defined to z/OS Communications Server as shared, and specify the **OPTLOCAL** keyword*



- WLM (Workload Manager) health value – IMS Connect
 - Automatically report the overall health of IMS Connect

- Allows Sysplex Distributor to use the information for workload balancing by coding a distribution method of SERVERWLM

VIPADYNAMIC ...

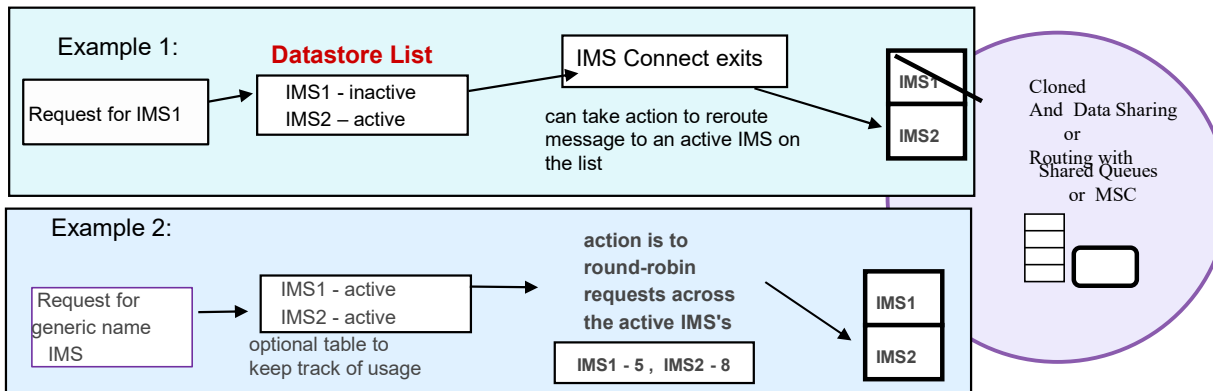
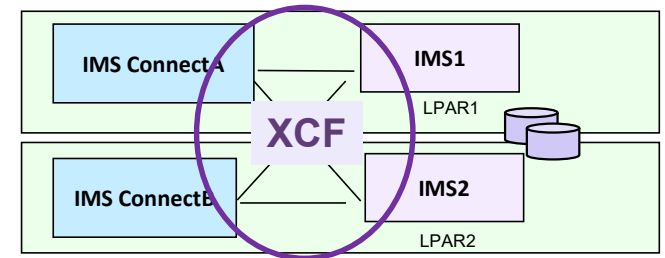
VIPADISTRIBUTE DEFINE DISTMETHOD **SERVERWLM**

Note: the health value is not used when DISTMETHOD is ROUNDROBIN or BASEWLM

And... With Expanding Environments

— IMS Connect – IMS

- IMS Connect supports workload balancing across IMS systems that can process the workload
 - Assumes the target IMS systems can process the workload (data sharing, shared queues, cloned systems....)
- When a message destination is resolved to a particular z/OS host and IMS Connect system
 - IMS Connect can access multiple IMS Systems (datastores)
 - Message exits can reroute a message to a different target IMS
 - For load balancing or if an IMS is unavailable
 - The Datastore table provides information as to which systems are active

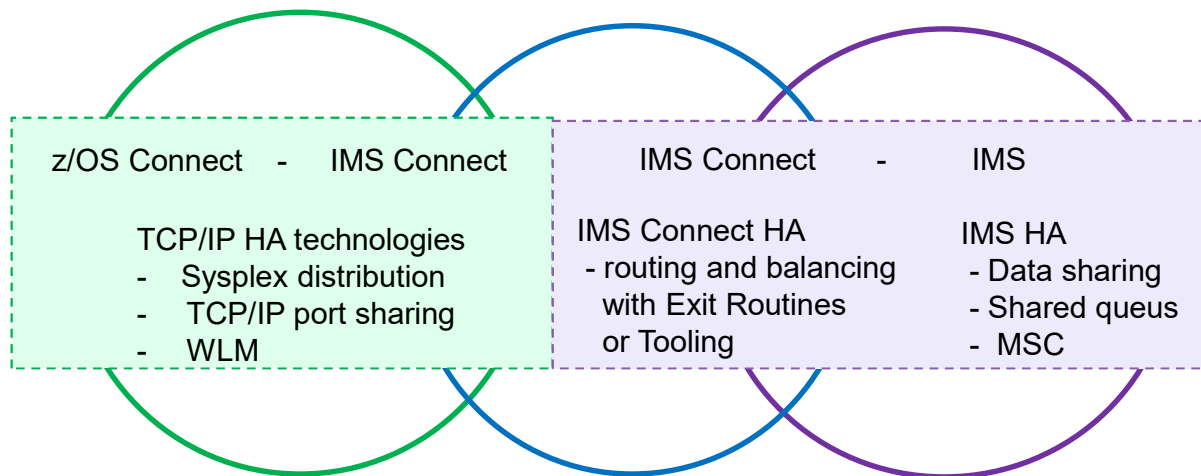


Tooling, e.g. IMS Connect Extensions (IMS CEX) enhances the functionality of IMS Connect for load balancing and rebalancing after a failure, routing, problem determination, etc.

- Without having to write exit routines

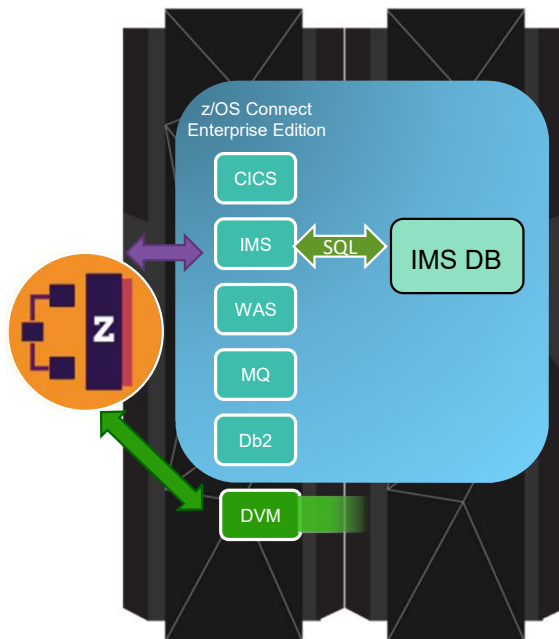
So... For High Availability and Resiliency

- Understand the requirements
- Understand what each connection environment in the path provides
 - For redundancy, failure and workload balancing



And Now... Access to IMS Databases

— API access through an IMS DB service provider



WHY?

Although moving data around has its uses



It is often BEST

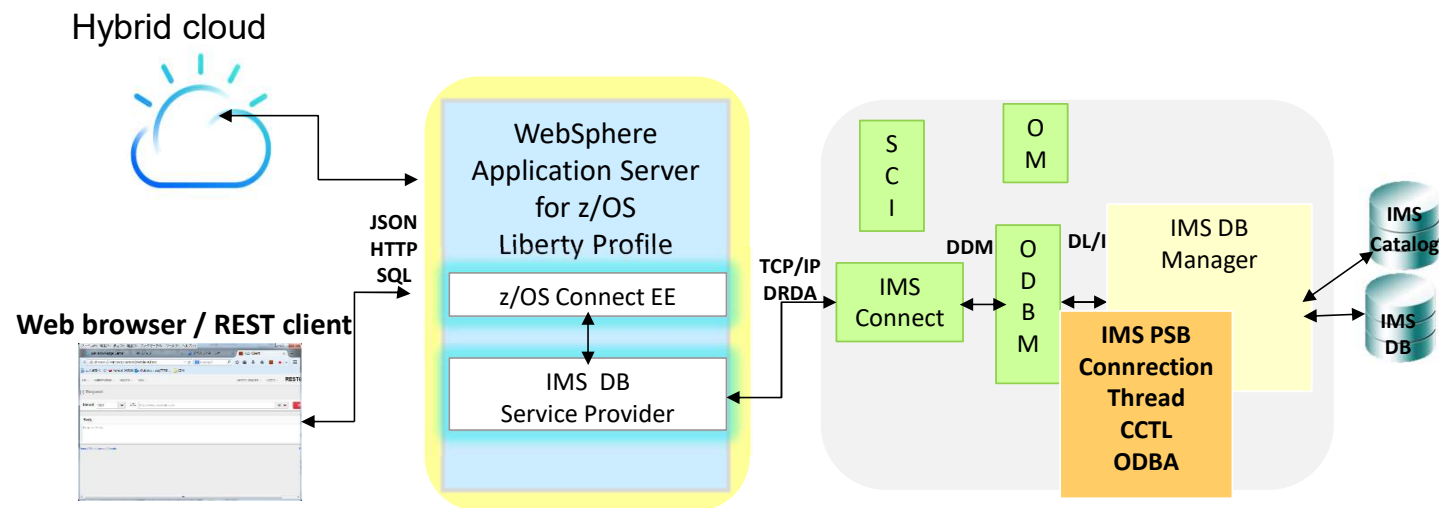


To keep data where it belongs

And where it is always fresh
(data can become stale as soon as it is moved)

IMS DB Service Provider

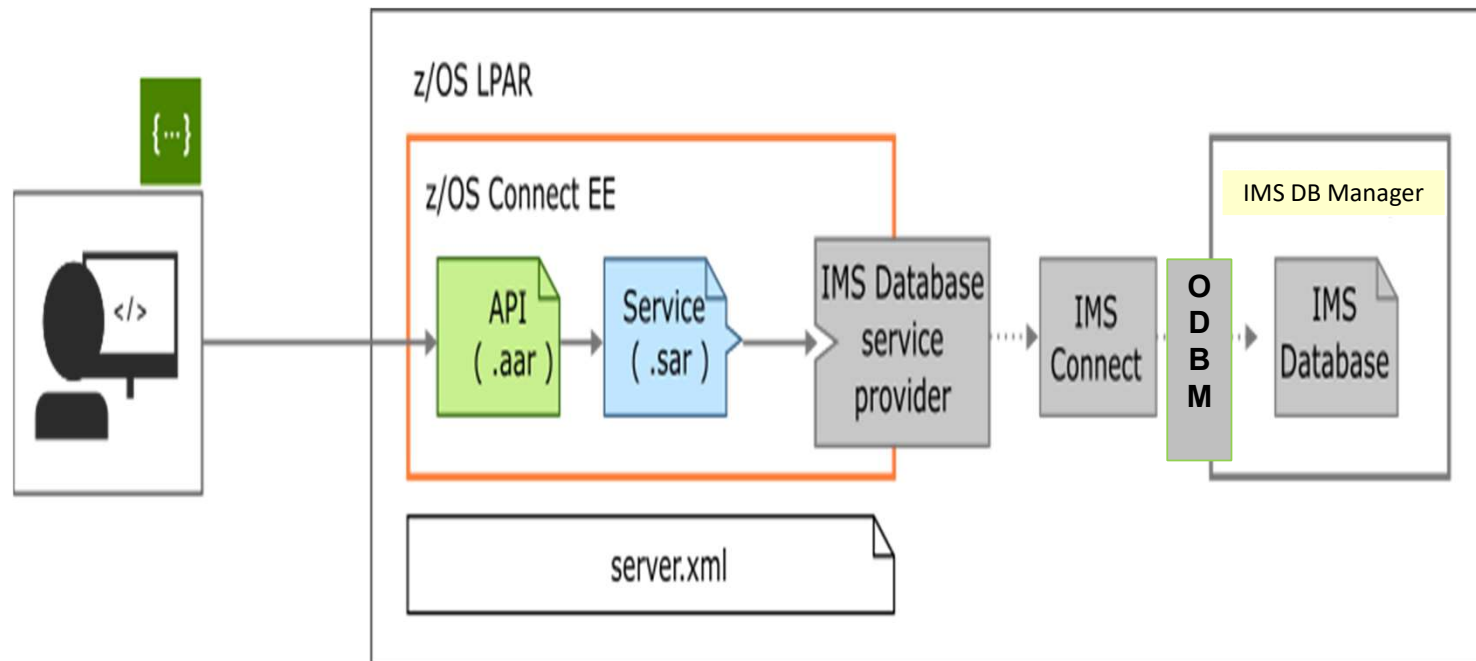
- A service provider for z/OS Connect EE that handles IMS Database requests:
 - Data transformation (JSON <-> byte array)
 - DRDA Interaction with IMS Connect and ODBM to access IMS databases
- z/OS Connect Enterprise Edition V3 includes the IMS DB Service Provider



Use Cases – Examples

- When applications need an SQL interface to IMS DB
- When applications need to retrieve specific IMS database records
- When existing IMS transactions cannot be API enabled.
- When you do not need complete database replication.
- You need to maintain Data Gravity and support Hybrid Cloud apps

Artifacts used for accessing IMS databases

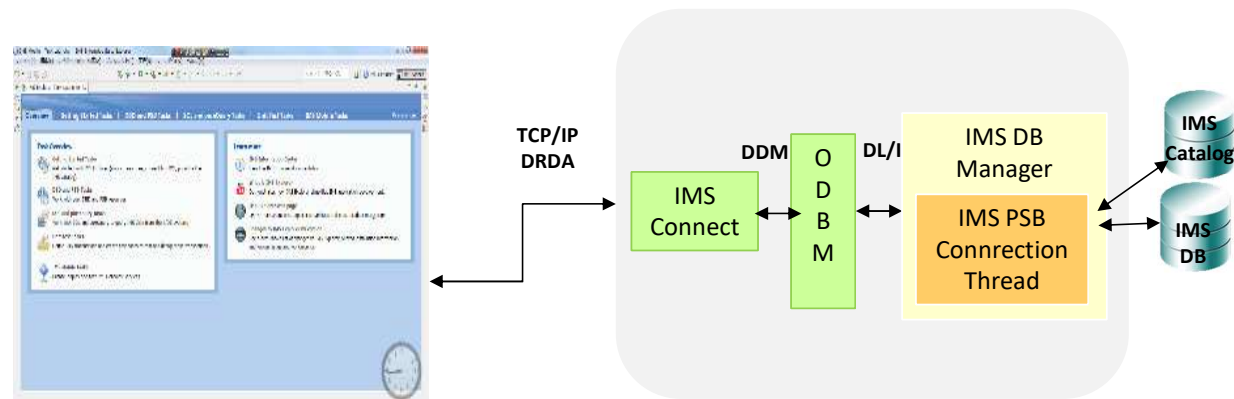


API archive (.aar) file

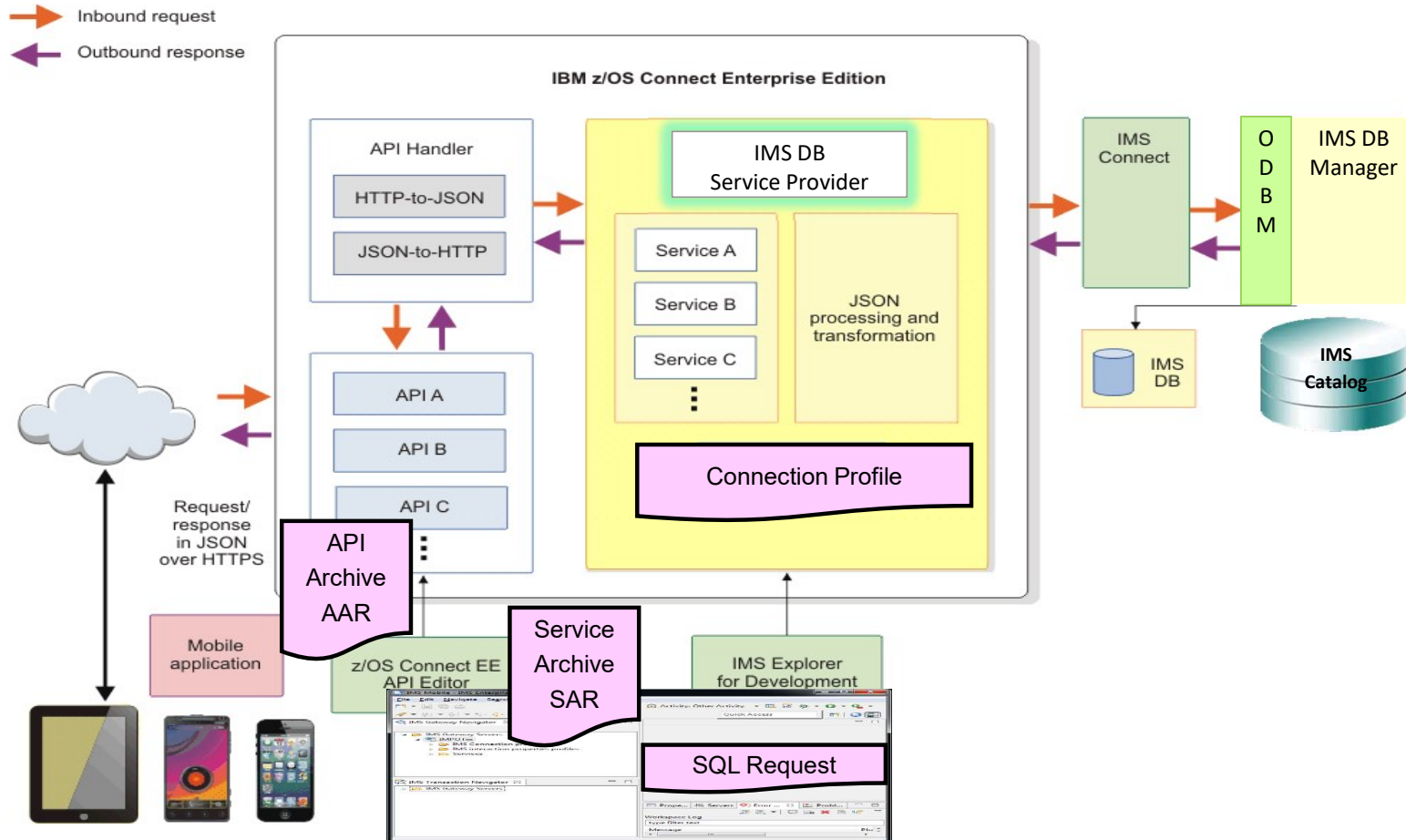
service archive (.sar) file

IMS Enterprise Suite Explorer for Development

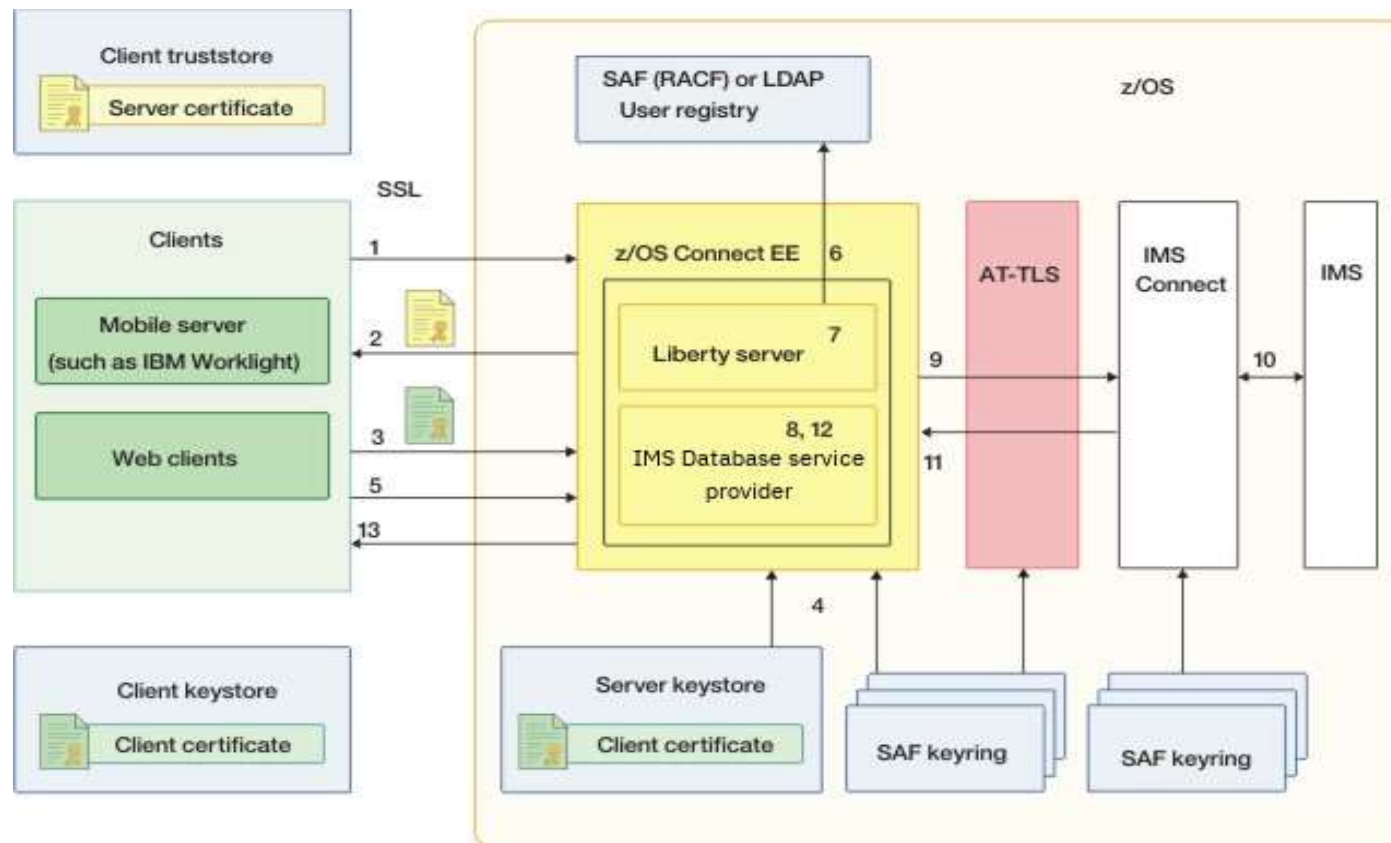
- Tooling for IMS DB service creation, testing, and management
 - Build SQL Queries
 - Test using DRDA Interaction with IMS Connect and ODBM to access IMS databases



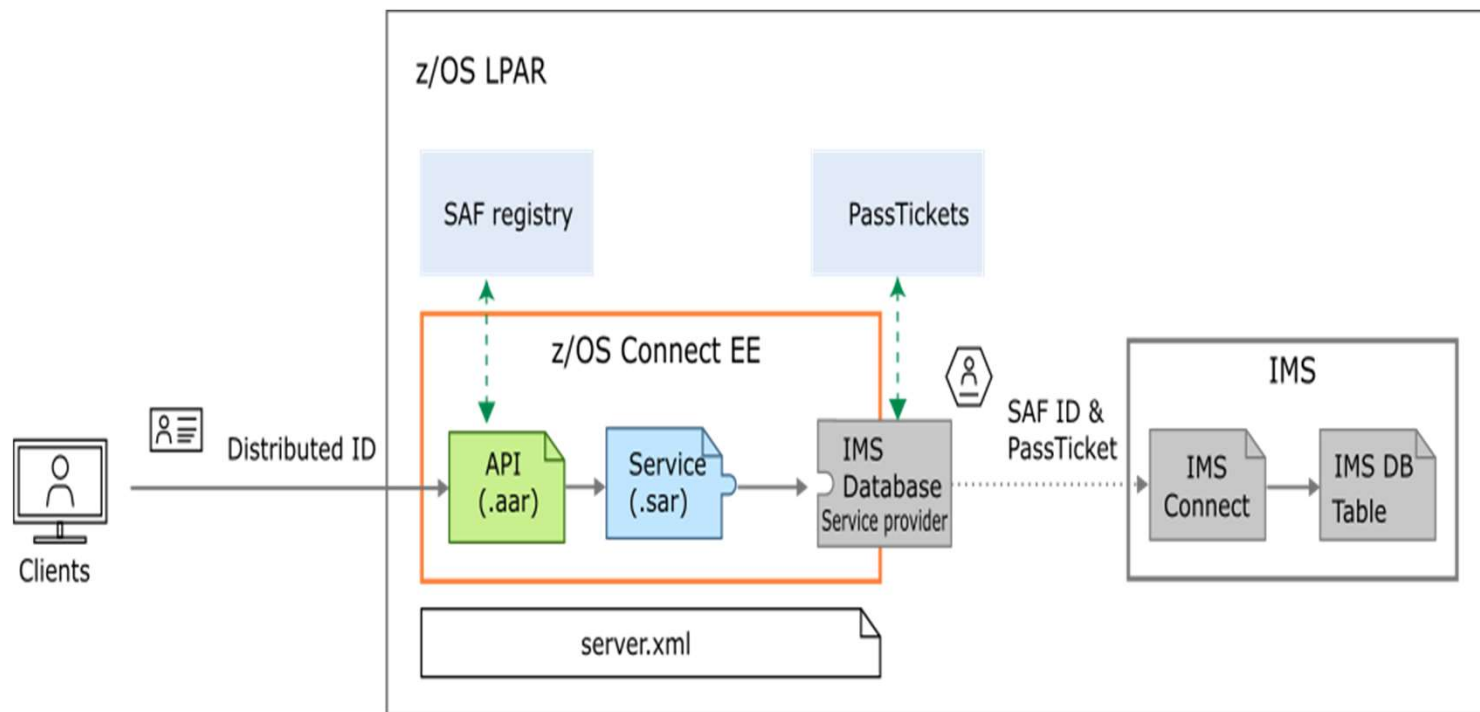
IMS DB Service Provider/z/OS Connect Enterprise Edition and IMS Explorer for Development



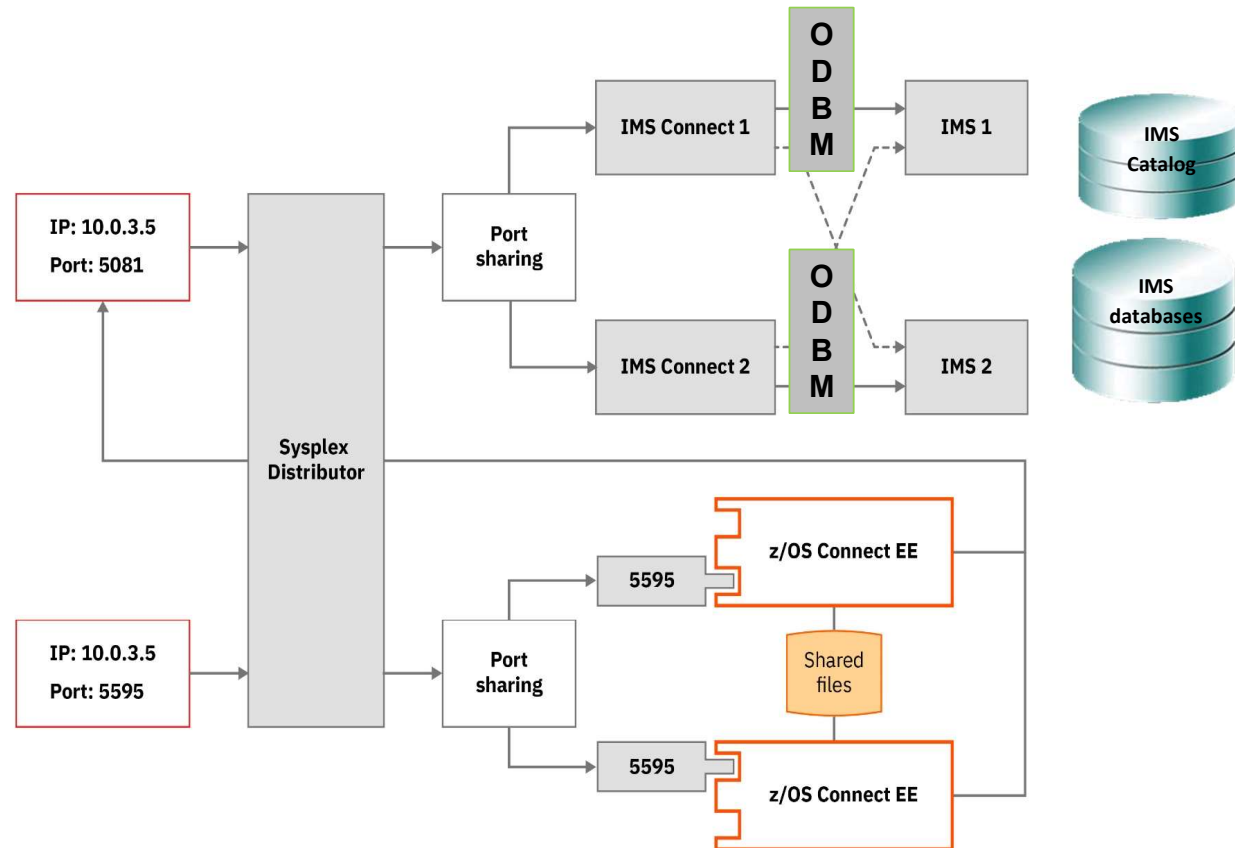
Security process flow and components involved



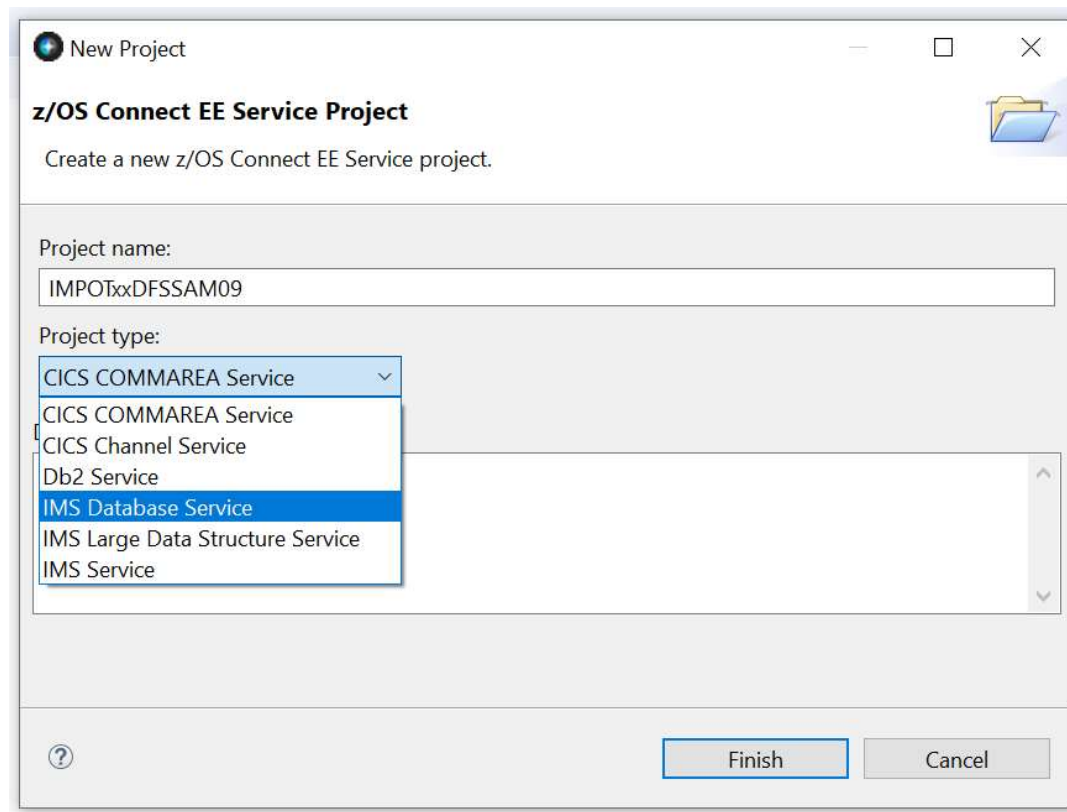
PassTickets Support



High Availability Configuration



z/OS Connect EE IMS Database Service Project



The image shows a 'New Project' dialog box with the title 'z/OS Connect EE Service Project'. It contains a text field for 'Project name' with the value 'IMPOTxxDFSSAM09' and a dropdown menu for 'Project type'. The dropdown menu is open, showing a list of project types: 'CICS COMMAREA Service', 'CICS COMMAREA Service', 'CICS Channel Service', 'Db2 Service', 'IMS Database Service' (which is highlighted), 'IMS Large Data Structure Service', and 'IMS Service'. At the bottom of the dialog are buttons for '?', 'Finish', and 'Cancel'.

New Project

z/OS Connect EE Service Project

Create a new z/OS Connect EE Service project.

Project name:
IMPOTxxDFSSAM09

Project type:
CICS COMMAREA Service
CICS COMMAREA Service
CICS Channel Service
Db2 Service
IMS Database Service
IMS Large Data Structure Service
IMS Service

? Finish Cancel

z/OS Connect EE IMS Database Service Project

zosexplorer - z/OS Connect Enterprise Edition - IMPOTxxDFSSAM09/service.properties - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Project Explorer

- IMPOTxxDFSSAM09
 - IMSDbServerAPIDFSSAM09
 - IMSDbServerDFSSAM09

z/OS Connect EE Servers Remote Systems

- New Connection
- Local
 - Local Files
 - Local Shells
- zserveros.centers.ihost.com
 - z/OS UNIX Files
 - z/OS UNIX Shells
 - MVS Files
 - TSO Commands
 - JES

IMPOTxxDFSSAM09 Service

Service Project Editor: Definition 4 errors detected

General Information

Edit or update the general information of the service.

Type: IMS Database Service

Version: 1.0.0

Description:

Actions

Steps to create a service:

1. Input service version.
2. Specify the SQL command for the service.
3. Specify Database Connection Properties and Generate Service Interface.
4. [Complete the configuration for the service.](#)
5. [Deploy the service.](#)
6. [Export the service.](#)

Enter or import SQL Command

SQL Command:

Import...

Definition Configuration

Host Connections Properties Progress Problems

Connections

type filter text

Add...

Edit...

Remove

Connect

Disconnect

Credentials

dnet545 [dnet545]

impot01 [IMPOT01]

Add...

Edit...

Remove

IMS Explorer SQL Query Builder

Build and Run

workspace - IMS Explorer - IMPOTxxDFSSAM09/SelectScript1.sql - IMS Enterprise Suite Explorer

File Edit Navigate Search Project SQL Run IMS Explorer Window Help

Project Explorer

- autodealer
- colonial
- dfssam09catalogimbsb
- IMPOTxxDFSSAM09
 - IMS Universal Drivers Metadata
 - JRE System Library [IMS Explorer]
 - Referenced Libraries
 - DBD
 - PSB
 - ImportDataStructuresManager.wfm
 - SelectScript1.sql

Data Source Explorer

- Database Connections
 - Derby Sample Connection
 - IMSA DB
 - IMSA DB 7016 DFSCP000
 - IMSB DB
 - IMSB DB
 - Authorization IDs
 - Catalogs
 - DFSCP000
 - DFSSAM09
 - IMSB DB 7013 DFSCP000
 - IMSC DB
 - IMSC DB 7011 DFSCP000
 - IMSD CAT 7001 DFSCP000
 - IMSD DB
 - IMSD DB 7001 DFSCP000

ImportDataStructuresManager.wfm

SelectScript1.sql

```
SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY,
PCB01.STOKSTAT.STOCKEY
FROM PCB01.PARTROOT, PCB01.STOKSTAT
```

PARTROOT

STOKSTAT

PARTKEY

PARTROOT_PARTKEY

STOCKEY

Run SQL

Connections

type filter text

Add...

Edit...

Remove

Connect

Disconnect

Db2 Service Manager

Database

(3)

IS FTP

IS Remote System (3)

SMF

Connect Enterprise Edition (2)

SERVEROS:23628 [DNET545]

SERVEROS:33628 [DNET545]

Credentials

DNET545 [impot01]

IMPOTxx [IMPOTxx]

Add...

Edit...

Remove

Properties Search Results Servers Error Log Problems SQL Results

Type query expression here

Status Result1

Status	Operat...	Date	Conne...	PARTROOT_PARTKEY	PARTKEY	PARTROC
✓ Succ		4/1/20...	IMSB D...	02CK05CW181K	02CK0...	02CK05C
✗ Fail		4/2/20...	IMSD DB	02CK05CW181K	02CK0...	02CK05C
✓ Succ		4/2/20...	IMSD DB	02CK05CW181K	02CK0...	02CK05C
✓ Succ SELECT...		4/4/20...	IMSD DB			
✓ Succ SELECT...		4/5/20...	IMSB DB			

Total 148 records shown

IMS Explorer SQL Query Builder

Translate SQL to DL/I to validate call pattern

/SelectScript1.sql - IMS Enterprise Suite Explorer

MS Explorer Window Help

ImportDataStructuresManager.wfm SelectScript1.sql IMS Explorer Task Launcher

```
SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY,
PCB01.STOKSTAT.STOCKEY
FROM PCB01.PARTROOT, PCB01.STOKSTAT
```

PARTROOT

☒ PARTKEY

☐ DISTINCT

Column	Conditions	Group
PCB01.PARTROOT.P...		A
PCB01.STOKSTAT.PA...		
PCB01.STOKSTAT.ST		

SQL to DL/I Call Translation

Connection profile: IMSB DB

Name of PSB to query: DFSSAM09

SQL statements: SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY, Y.

DL/I calls:

```
GU  PARTROOT*D
    STOKSTAT

(LOOP)
GN  PARTROOT*D
    STOKSTAT
```

Translate

IMS Explorer SQL Query Builder

Copy SQL

workspace - IMS Explorer - IMPOTxxDFSSAM09/SelectScript1.sql - IMS Enterprise Suite Explorer

File Edit Navigate Search Project SQL Run IMS Explorer Window Help

Project Explorer

- autodealer
- colonial
- dfssam09catalogimsb
- IMPOTxxDFSSAM09
 - IMS Universal Drivers Metadata
 - JRE System Library [IMS Explorer]
 - Referenced Libraries
 - DBD
 - PSB
 - ImportDataStructuresManager.wfm
 - SelectScript1.sql

Data Source Explorer

- Database Connections
 - Derby Sample Connection
 - IMSA DB
 - IMSA DB 7016 DFSCP000
 - IMSB DB
 - Authorization IDs
 - Catalogs
 - DFSCP000
 - DFSSAM09
 - IMSB DB 7013 DFSCP000
 - IMSC DB
 - IMSC DB 7011 DFSCP000
 - IMSD CAT 7001 DFSCP000
 - IMSD DB
 - IMSD DB 7001 DFSCP000

ImportDataStructuresManager.wfm *SelectScript1.sql

```
SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY,
       PCB01.STOKSTAT.STOCKEY
FROM PCB01.PARTROOT, PCB01.STOKSTAT
```

PARTROOT PARTKEY STOKSTAT PARTROOT STOCKEY

Columns Conditions Groups

Column	Alias	Out...	Sort Type	Sort Or...
PCB01.PARTROOT.P...		☒		
PCB01.STOKSTAT.PA...		☒		
PCB01.STOKSTAT.ST...		☒		

Properties Search Results Servers Error Log Problems SQL Results

Type query expression here Status Result1

Status	Operat...	Date	Conne...
✖ Failer		4/2/20...	IMSD DB
✔ Succ		4/2/20...	IMSD DB
✔ Succ SELECT...		4/4/20...	IMSD DB
✔ Succ SELECT...		4/5/20...	IMSB DB
✔ Succ SELECT...		4/5/20...	IMSB DB

Elapsed Time: 0 hr, 0 min, 4 sec, 990 ms.

Connections

type filter text Add... Edit... Remove Connect Disconnect

- b2 Service Manager
- MS Database
- OS (3)
- z/OS FTP
- z/OS Remote System (3)
- z/OSMF
- OS Connect Enterprise Edition (2)
- ZSERVEROS:23628 [DNET545]
- zserveros:33628 [DNET545]

Credentials

DNET545 [impot01] Add... Edit... Remove

IMPOTxx [IMPOTxx] Add... Edit... Remove

z/OS Connect EE IMS Database Service Project

Paste SQL

zosexplorer - z/OS Connect Enterprise Edition - IMPOTxxDFSSAM09/service.properties - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Project Explorer

- services
 - IMSDbserverDFSSAM09
 - files
 - interfaces
 - request.si
 - response.si
 - service
 - IMSDbserverDFSSAM09.xml
 - schemas
 - IMSDbserverDFSSAM09Request.json
 - IMSDbserverDFSSAM09Response.json
 - manifest.yml

z/OS Connect EE Servers Remote Systems

Type to filter by name or status

- impot01zserveros.centers.ihost.com (zserveros.centers.ihost.com:33628)
- zserveros:23628 (zserveros.centers.ihost.com:23628)
 - APIs (19)
 - Services (20)
- zserveros:33628 (zserveros.centers.ihost.com:33628)

Service Project Editor: Definition 3 errors detected

General Information

Edit or update the general information of the service.

Type: IMS Database Service

Version: 1.0.0

Description:

Actions

Steps to create a service:

1. Input service version.
2. Specify the SQL command for the service.
3. Specify Database Connection Properties and Generate Service Interface.
4. [Complete the configuration for the service.](#)
5. [Deploy the service.](#)
6. [Export the service.](#)

Enter or import SQL Command

SQL Command: SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY, PCB01.STOKSTAT.STOCKEY FROM PCB01.PARTROOT, PCB01.STOKSTAT

Import...

Definition Configuration

Host Connections Properties Progress Problems

Connections

type filter text

Add... Edit... Remove Connect Disconnect

- Db2 Service Manager
- IMS Database (1)
 - DBserverzserveros:7001 [impot01]
- z/OS (1)
 - z/OS FTP
- z/OS Remote System (1)

Credentials

Add... Edit... Remove

- dnet545 [dnet545]
- impot01 [IMPOT01]

z/OS Connect EE IMS Database Service Project Generate Service

zosexplorer - z/OS Connect Enterprise Edition - IMPTxxDFSSAM09/service.properties - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Quick Access

Project Explorer

- services
 - IMSDbserverDFSSAM09
 - files
 - interfaces
 - request.si
 - response.si
 - service
 - IMSDbserverDFSSAM09.xml
 - schemas
 - IMSDbserverDFSSAM09Request.json
 - IMSDbserverDFSSAM09Response.json
 - manifest.yml
 - package.xml

z/OS Connect EE Servers Remote Systems

Type to filter by name or status

- impot01zserveros.centers.ihost.com (zserveros.centers.ihost.com:2362R)
- zserveros:2362R (zserveros.centers.ihost.com:2362R)

*IMPTxxDFSSAM09 Service

Edit or update the general information of the service.

Type: IMS Database Service

Version: 1.0.0

Description:

Steps to create a service:

1. Input service version.
2. Specify the SQL command for the service.
3. Specify Database Connection Properties and Generate Service Interface.
4. [Complete the configuration for the service.](#)
5. [Deploy the service.](#)
6. [Export the service.](#)

Generate Service Interface

Service Interface was generated successfully.

OK

Specify Database Connection Properties and Generate Service Interface

Database Connection: IMSB Database Name: DFSSAM09 Generate Service Interface...

z/OS Connect EE IMS Database Service Project Deploy Service Archive file(SAR)

zosexplorer - z/OS Connect Enterprise Edition - IMPOTxxDFSSAM09/service.properties - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Project Explorer

- IMPOTxxDFSSAM09
 - bin
 - schemas
 - IMPOttxd
 - IMPOttxd
 - IMPOttxDF
 - service-interfa
 - request.si
 - response.si
 - service.proper
 - IMSDbServerAP
 - IMSDbServerDF

IMPOTxxDFSSAM09 Service

Steps to create a service:

1. Input service version.
2. Specify the SQL command for the service.
3. Specify Database Connection Properties and Generate Service Interface.
4. [Complete the configuration for the service.](#)
5. [Deploy the service.](#)
6. [Export the service.](#)

Import SQL Command

and: SELECT PCB01.PARTROOT.PARTKEY, PCB01.STOKSTAT.PARTROOT_PARTKEY, PCB01.STOKSTAT.STOCKEY FROM PCB01.PARTROOT, PCB01.STOKSTAT

Database Connection Properties and Generate Service Interface

Connection: IMSB Database Name: DFSSAM09 Generate Service Interface...

Configuration

Actions Properties Progress Problems

z/OS Connect EE

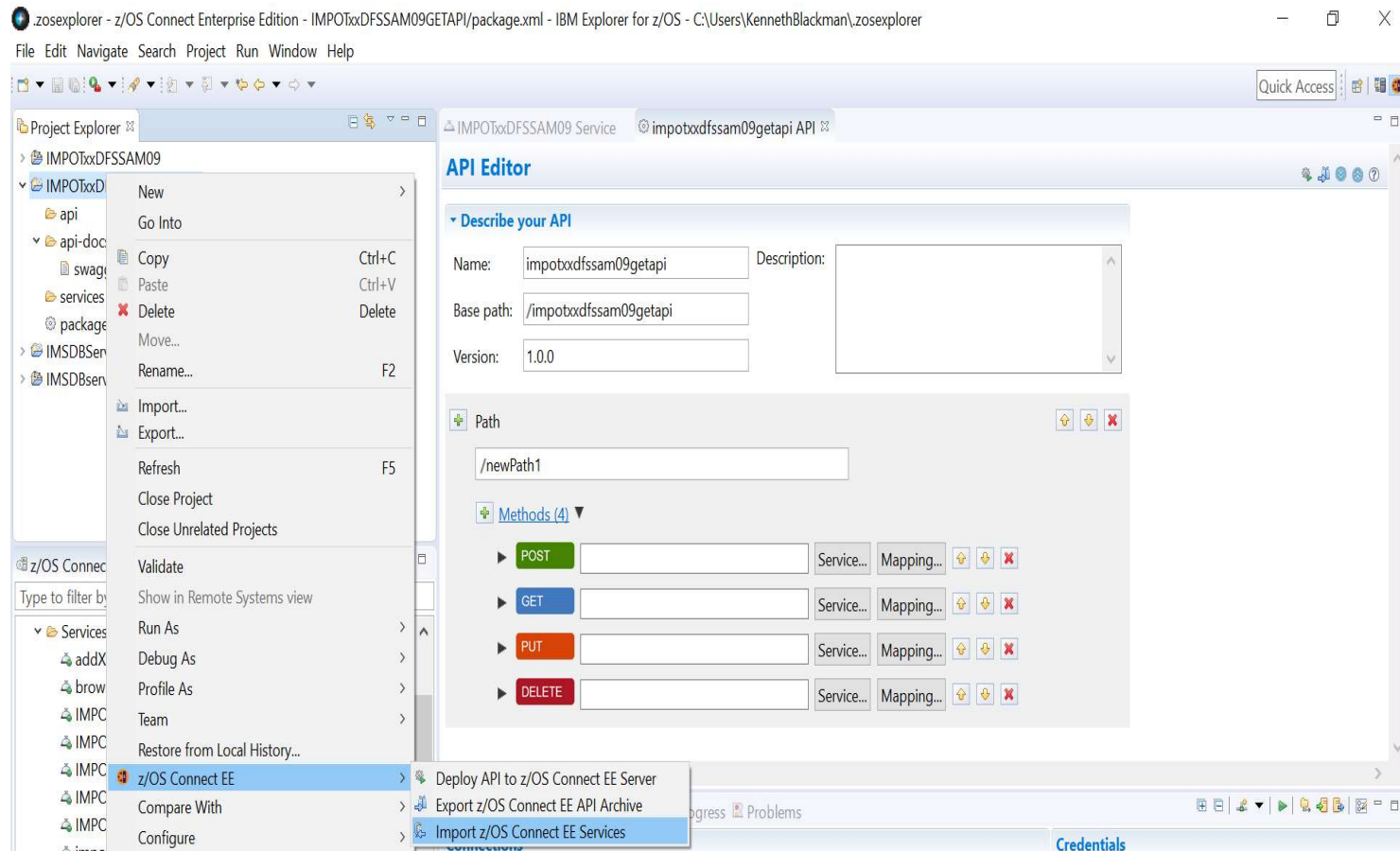
- Deploy Service to z/OS Connect EE Server
- Export z/OS Connect EE Service Archive

Credentials

Username	Password	Action
dnet545 [dnet545]		Add...
impot01 [IMPOT01]		Edit...
IMPOTxx [IMPOT01]		Remove

z/OS Connect EE IMS Database Service Project

Create API using the SAR file



z/OS Connect EE IMS Database Service Project

Create POST API

zosexplorer - z/OS Connect Enterprise Edition - IMPOTxxDFSSAM09GETAPI/package.xml - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Project Explorer

- IMPOTxxDFSSAM09
 - IMPOTxxDFSSAM09GETAPI
 - api
 - api-docs
 - swagger.json
 - services
 - package.xml
 - IMSDbServerAPIDFSSAM09
 - IMSDbServerDFSSAM09

z/OS Connect EE Servers Remote Systems

Type to filter by name or status

- impot01zserveros.centers.ihost.com (zserveros.centers.ihost.com)
- zserveros:23628 (zserveros.centers.ihost.com:23628)
- zserveros:33628 (zserveros.centers.ihost.com:33628)
- zserverosIMPOTxx (zserveros.centers.ihost.com:23628)
 - APIs (19)
 - Services (21)
 - addXX (Started)
 - browseXX (Started)
 - IMPOT01IVTNO01 (Started)
 - IMPOT03IVTNO3 (Started)
 - IMPOT04IVTNO04 (Started)
 - IMPOT07IVTNO07 (Started)

IMPOTxxDFSSAM09 Service

impotxxdfssam09getapi API

API Editor

Describe your API

Name: impotxxdfssam09getapi Description:

Base path: /impotxxdfssam09getapi

Version: 1.0.0

Path

/newPath1

Methods (1)

POST IMPOTxxDFSSAM09 Service... Mapping...

Host Connections Properties Progress Problems

Connections

type filter text

Add... Edit... Remove Connect

- Db2 Service Manager
- IMS Database (2) (IMSB)
 - DBserverzserveros:7001 [impot01]
 - IMSB [IMPOTxx]

Credentials

Add... Edit... Remove

- dnet545 [dnet545]
- impot01 [IMPOT01]
- IMPOTxx [IMPOT01]

z/OS Connect EE IMS Database Service Project

Deploy POST API

The screenshot displays the IBM z/OS Connect EE IDE interface. The top menu bar includes File, Edit, Navigate, Search, Project, Run, Window, and Help. The Project Explorer on the left shows the project structure for IMPOTxxDFSSAM09, including folders for api, newPath1, POST, mapping, api-docs, swagger.json, services, IMPOTxxDFSSAM09, files, schemas, and manifest.yml. A context menu is open over the POST folder, listing actions such as New, Go Into, Copy, Paste, Delete, Move, Rename, Import, Export, Refresh, Close Project, Close Unrelated Projects, Validate, Show in Remote Systems view, Run As, Debug As, Profile As, Team, Restore from Local History, and z/OS Connect EE. The z/OS Connect EE option is highlighted, and a sub-menu is visible with options: Deploy API to z/OS Connect EE Server, Export z/OS Connect EE API Archive, Import z/OS Connect EE Services, Db2 Service Manager, IMS Database (2) (IMSB), DBserverzserveros:7001 [impot01], and IMSB [IMPOTxx]. The API Editor on the right shows the configuration for the POST API, including the name (impotxxdfssam09getapi), description, base path (/impotxxdfssam09getapi), version (1.0.0), and path (/newPath1). The Methods section shows a POST method associated with the IMPOTxxDFSSAM09 service. The bottom right panel displays the Credentials section with a list of credentials: dnet545 [dnet545], impot01 [IMPOT01], and IMPOTxx [IMPOT01], each with Add, Edit, Remove, and Connect buttons.

z/OS Connect EE IMS Database Service Project Test POST API

zosexplorer - z/OS Connect Enterprise Edition - http://zserveros.centers.ihost.com:23628/impotxxdfssam09getapi - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Quick Access

Project Explorer

- IMPOtXXDFSSAM09
 - IMPOtXXDFSSAM09GETAPI
 - api
 - newPath1
 - POST
 - mapping.xml
 - api-docs
 - swagger.json
 - services
 - IMPOtXXDFSSAM09
 - files
 - schemas
 - manifest.yml
 - package.xml

z/OS Connect EE Servers Remote Systems

Type to filter by name or status

APIs (20)

- addxxapi (Started)
- browsexxapi (Started)
- impot01ivtno01getapi (Started)
- impot04ivtno04getapi (Started)
- impotxxdfssam09getapi (Started)
- impotxxivtnoxxgetapi (Started)
- imsdblockchainapi (Started)
- imsdbserverapidfssam09 (Started)
- imsdchintran (Started)
- invtnoxxapi (Started)
- phoneadd (Started)

swagger

impotxxdfssam09getapi

impotxxdfssam09getapi

Show/Hide | List Operations | Expand Operations

[BASE URL: /impotxxdfssam09getapi , API VERSION: 1.0.0]

Host Connections Properties Progress Problems

Connections

type filter text

Add...

Edit...

Remove

Connect

Credentials

dnet545 [dnet545]

impot01 [IMPOT01]

IMPOTxx [IMPOT01]

Add...

Edit...

Remove

Done

z/OS Connect EE IMS Database Service Project Results

z/osexplorer - z/OS Connect Enterprise Edition - http://zserveros.centers.ihost.com:23628/impotxdfssam09getapi - IBM Explorer for z/OS - C:\Users\KennethBlackman\zosexplorer

File Edit Navigate Search Project Run Window Help

Project Explorer

- IMPOTxdfSSAM09
 - IMPOTxdfSSAM09GETAPI
 - api
 - newPath1
 - POST
 - mapping.xml
 - api-docs
 - swagger.json
 - services
 - IMPOTxdfSSAM09
 - files
 - schemas
 - manifest.yml
 - package.xml

z/OS Connect EE Servers Remote Systems

Type to filter by name or status

APIs (20)

- addxapi (Started)
- browsexapi (Started)
- impot01ivtno01getapi (Started)
- impot04ivtno04getapi (Started)
- impotxdfssam09getapi (Started)
- impotxivtnoxgetapi (Started)
- imsdblockchainapi (Started)
- imsdbserverapidfssam09 (Started)
- imsdchintrn (Started)
- invtnoxapi (Started)
- phoneadd (Started)

Response Body

```
{
  "response": {
    "result": [
      {
        "PARTKEY": "02AN960C10",
        "PARTROOT_PARTKEY": "02AN960C10",
        "STOCKKEY": "00 AA16511"
      },
      {
        "PARTKEY": "02AN960C10",
        "PARTROOT_PARTKEY": "02AN960C10",
        "STOCKKEY": "00 AK2877F"
      },
      {
        "PARTKEY": "02AN960C10",
        "PARTROOT_PARTKEY": "02AN960C10",
        "STOCKKEY": "0028009126"
      }
    ]
  }
}
```

Host Connections Properties Progress Problems

Connections

type filter text

- Db2 Service Manager
- IMS Database (2) (IMSB)
 - DBserverzserveros:7001 [impot01]
 - IMSB [IMPOTxx]

Add... Edit... Remove Connect

Credentials

- dnet545 [dnet545]
- impot01 [IMPOT01]
- IMPOTxx [IMPOT01]

Add... Edit... Remove

IMS Open Access Infrastructure

- **IMS version 15 IMSPLEX**
 - Structured Call Interface (SCI)
 - Operations Manager (OM)
 - Open Database Manager (ODBM)
 - IMS Connect
 - DRDA TCP/IP Port
 - IMS Control Region
 - IMS Catalog
 - Metadata for your IMS databases
- **IMS Explorer for Development**
 - Build and Test SQL queries
- **z/OS Connect EE**
 - WebSphere® Application Server for z/OS Liberty runtime
 - z/OS Connect EE API toolkit

IMS Service recommendations

- **IMS v15 PH15107/UI65936**
 - ensure ODBM availability for database services under heavy load
- **IMS v15 PH14651**
 - support the use of passphrases
- **IMS v15 PH19383**
 - provide additional support for user defined types.

So, In Summary... for IMS Resource (Transactions and Databases)

— As technology continues to evolve

- The IMS services and infrastructure
Extend and integrate IMS resources into these new environments
 - > To answer your needs for continued growth